

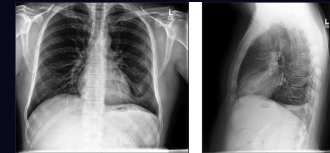
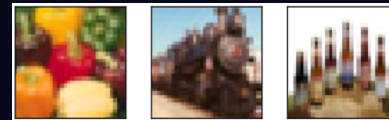
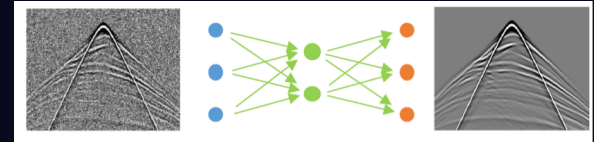
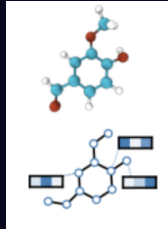
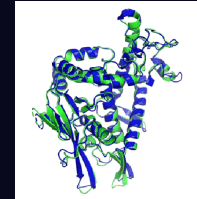
Lecture 3: Advanced Topics in Optimization

Dr Tiffany Vlaar
University of Glasgow

Using vs. Understanding Deep Learning

It can do a lot ...

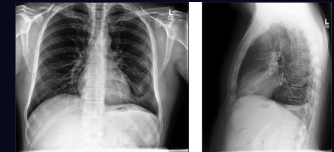
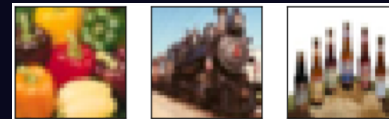
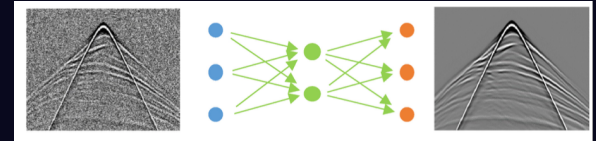
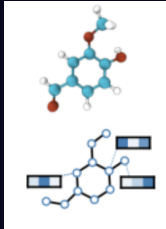
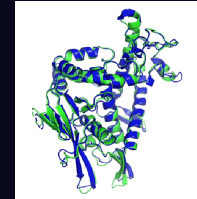
Deep Learning



Using vs. Understanding Deep Learning

It can do a lot ...

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TITLE

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YEAR

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2015

S Ioffe, C Szegedy

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Attention is all you need

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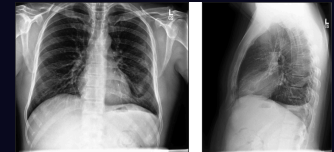
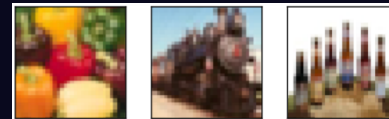
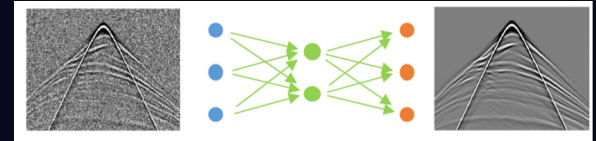
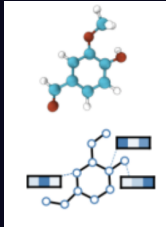
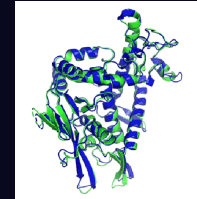
A Vaswani, N Shazeer, N Parmar, J Uszkoreit, L Jones, AN Gomez, ...

Despite widespread use, many open questions

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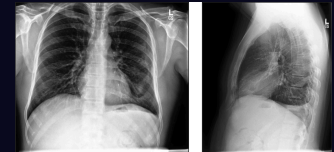
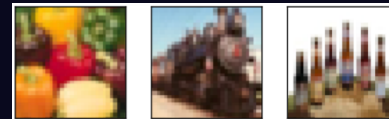
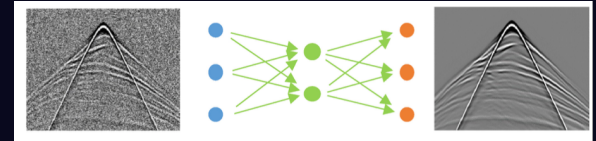
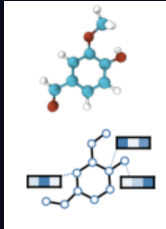
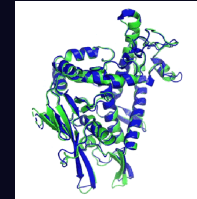
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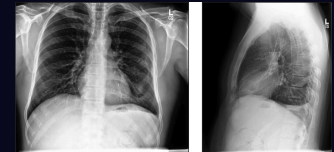
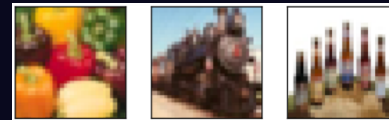
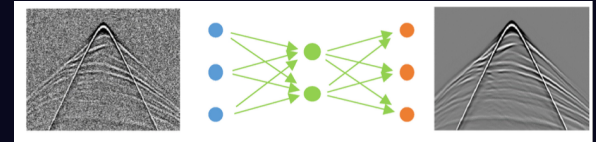
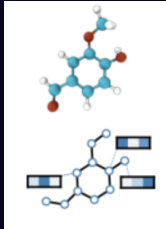
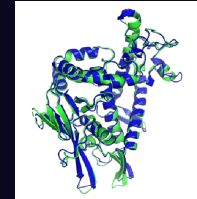
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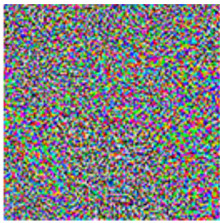
2017

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Further Reflections



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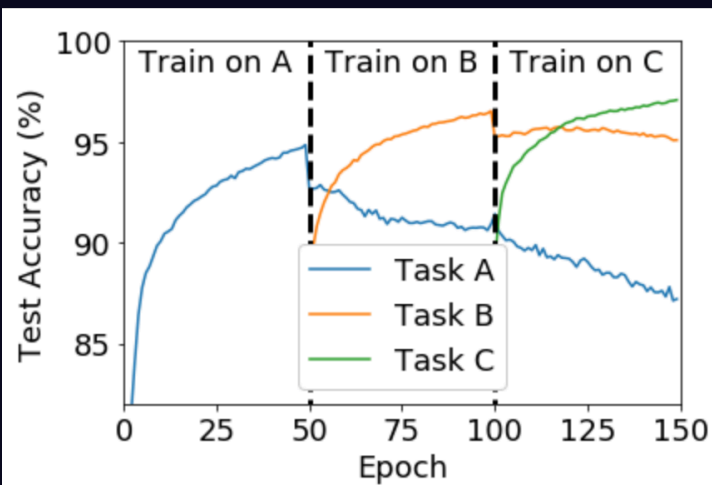
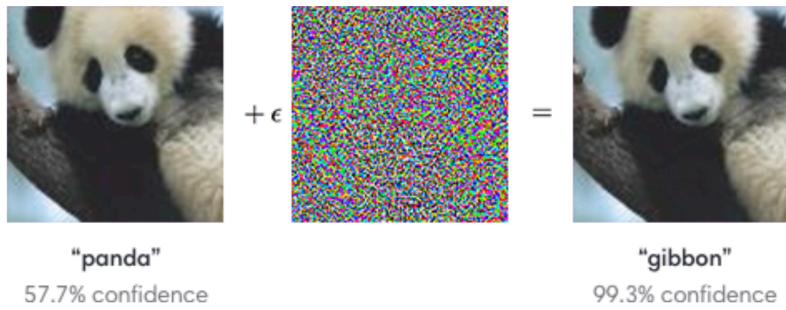
"panda"

57.7% confidence

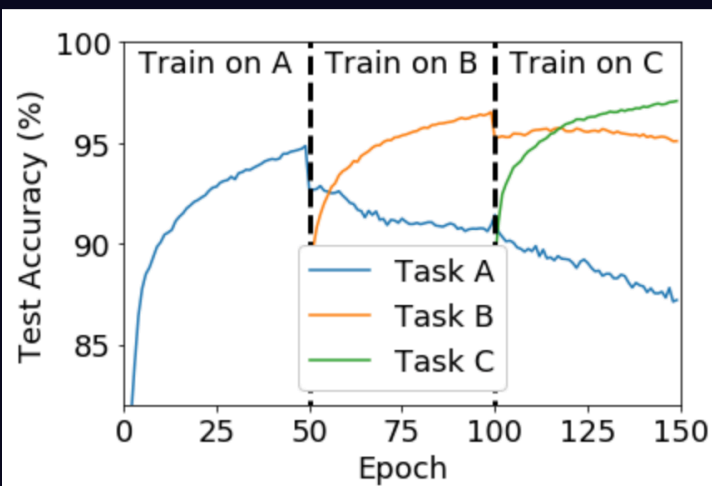
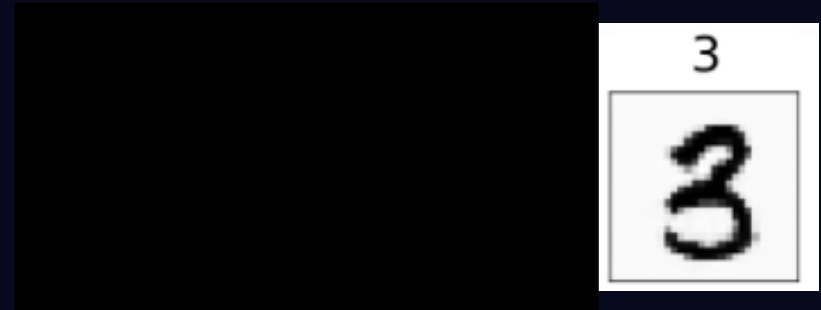
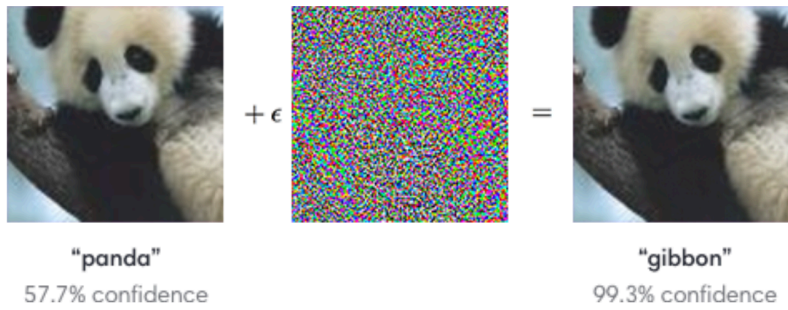
"gibbon"

99.3% confidence

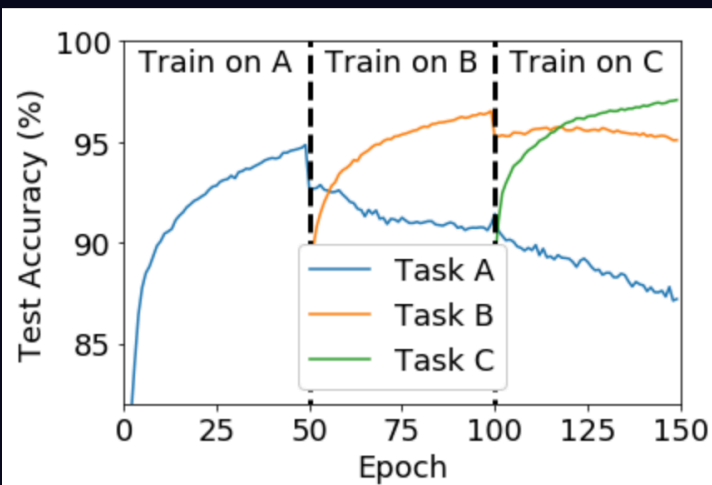
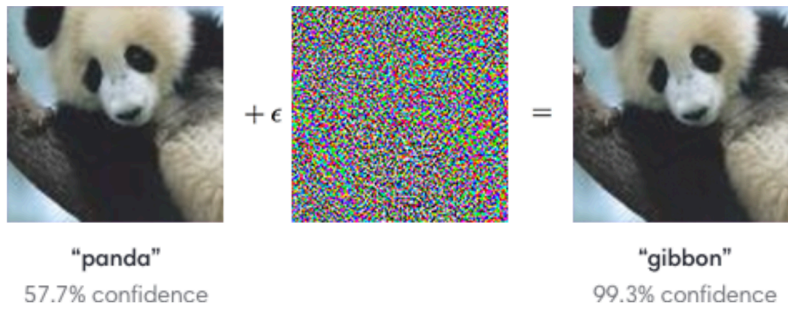
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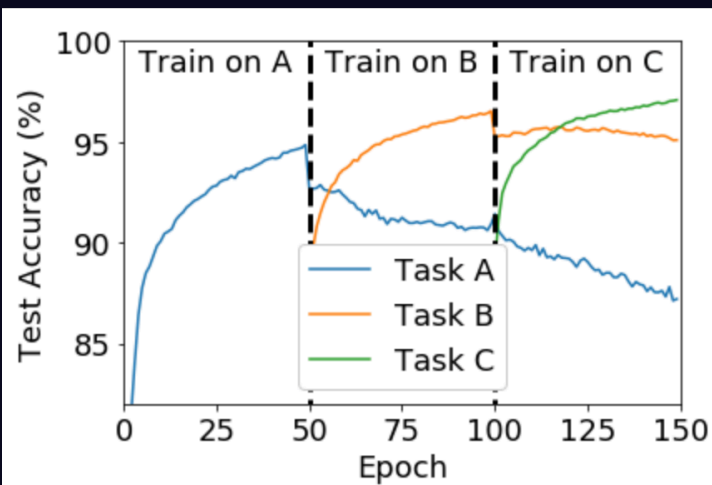
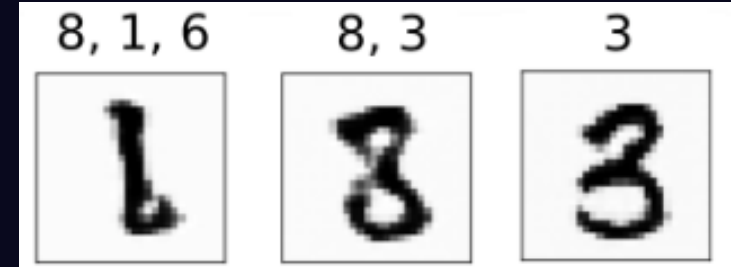
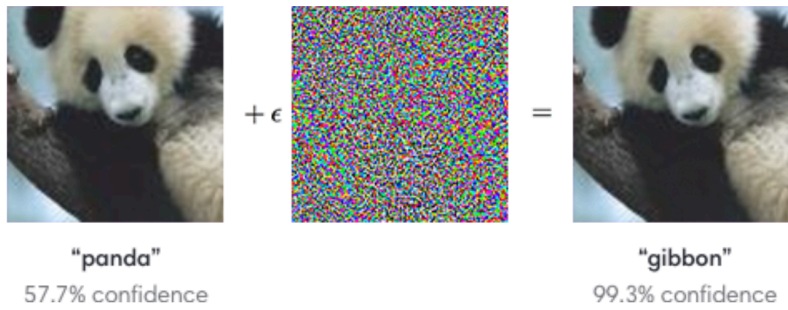
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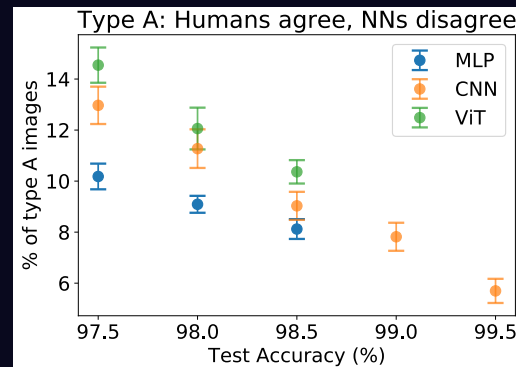
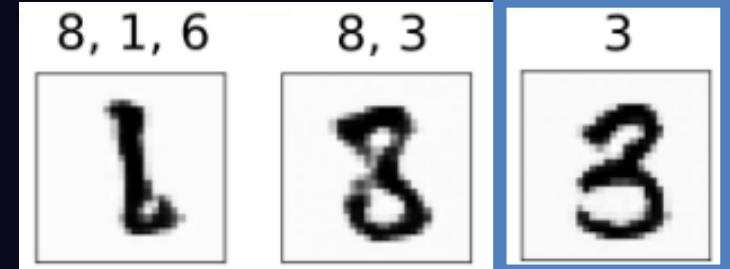
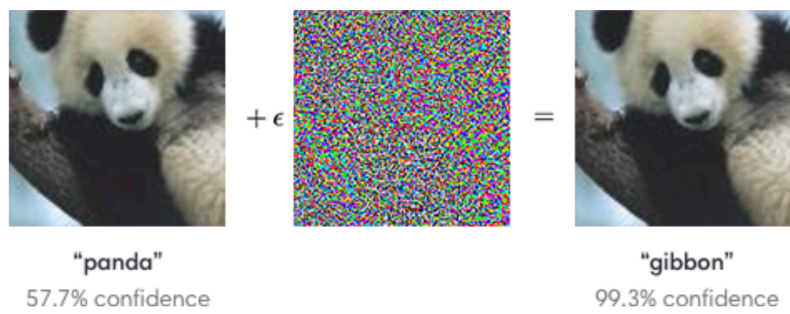
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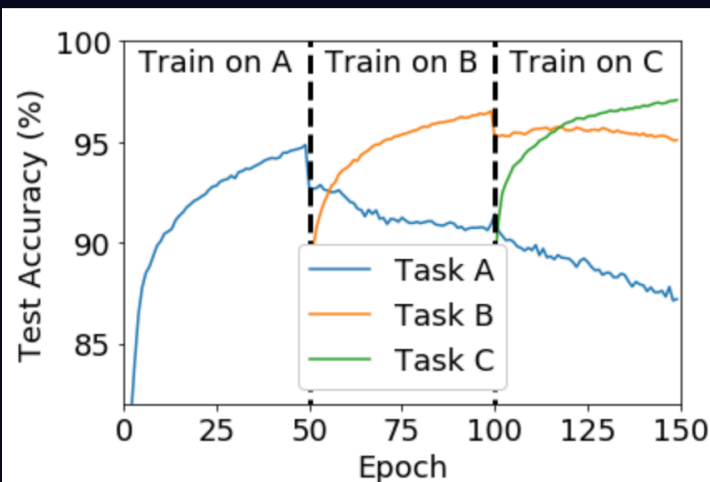
Further Reflections



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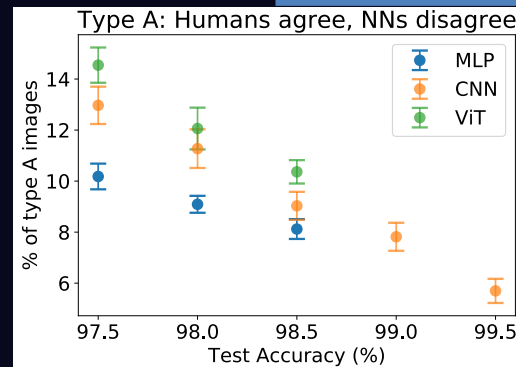
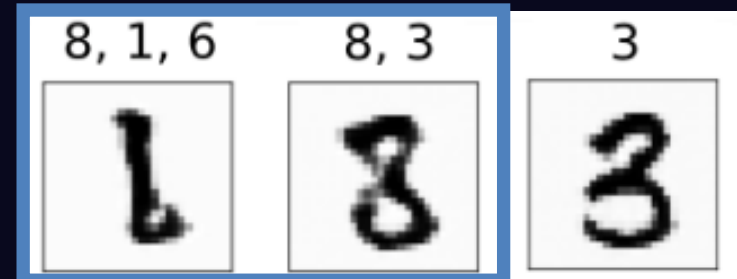
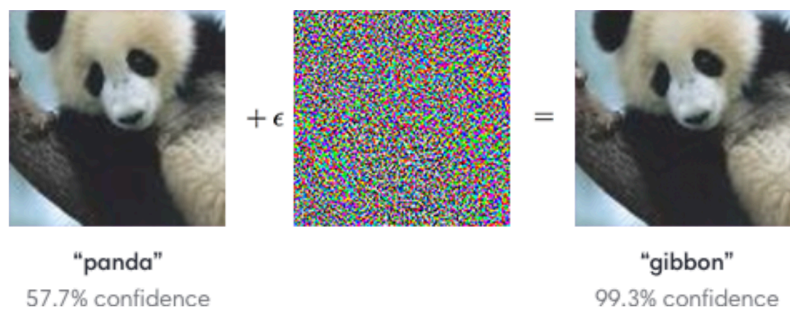


As test accuracy increases:
Misclassification decreases

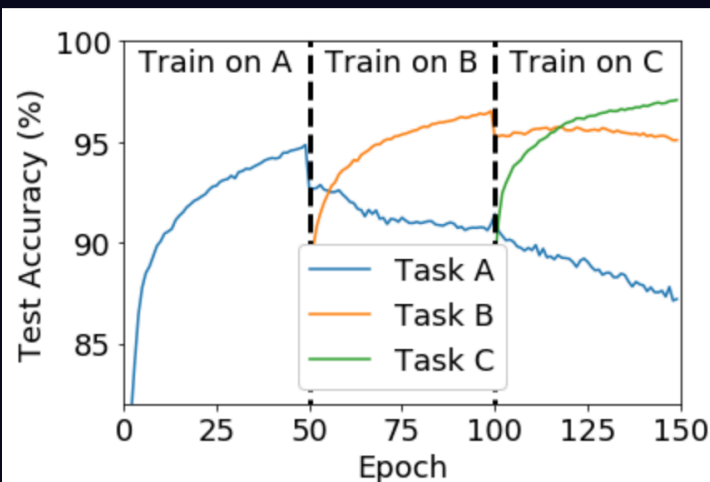


Kirkpatrick et al., 2017

Further Reflections

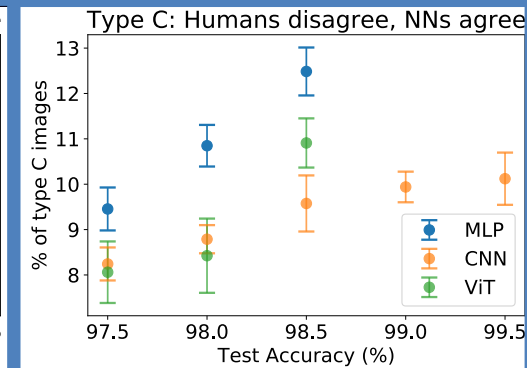
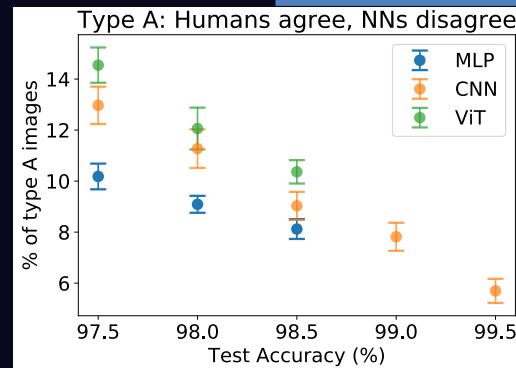
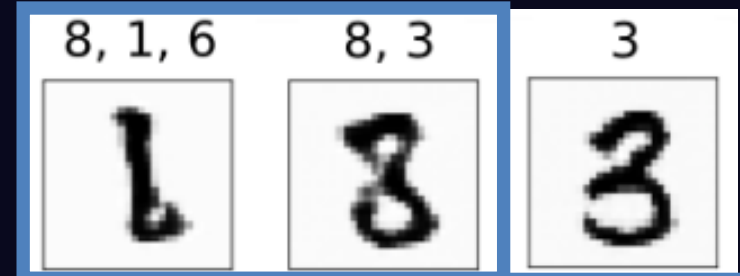
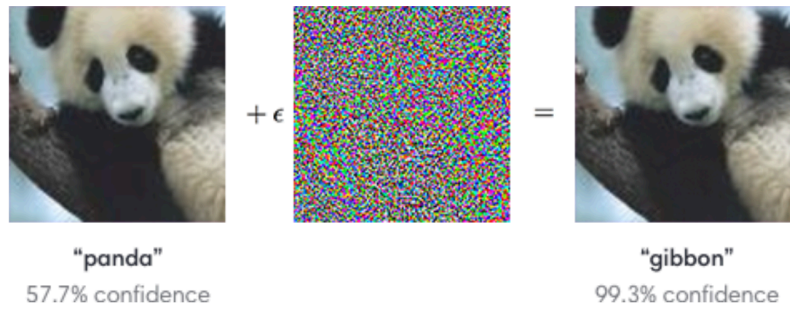


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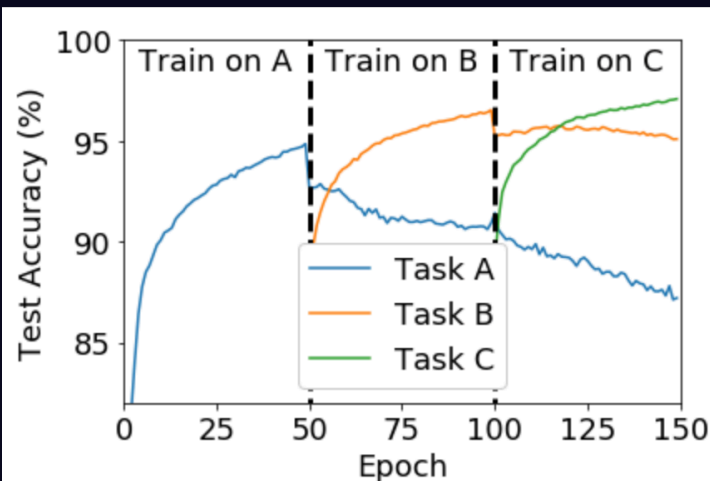
Kirkpatrick et al., 2017

Further Reflections



As test accuracy increases:
Misclassification decreases

Nets agree more on (for humans) difficult images

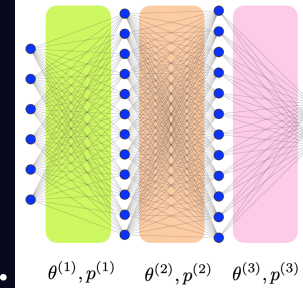


Kirkpatrick et al., 2017

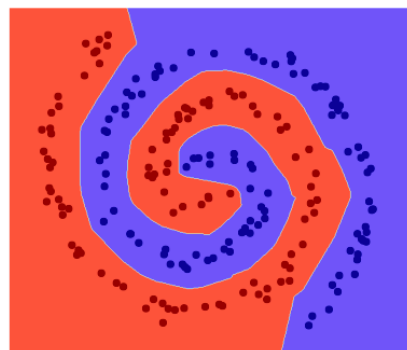
Try to enhance our understanding of neural networks

And use these findings to reduce computational cost

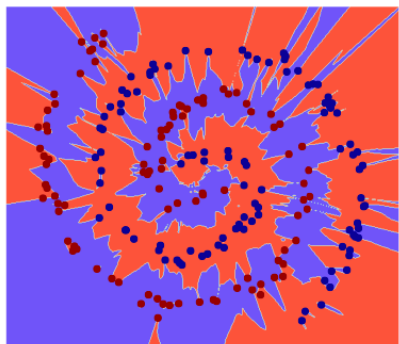
while enhancing or maintaining generalization.



Generalization



(a) 100% train, 100% test



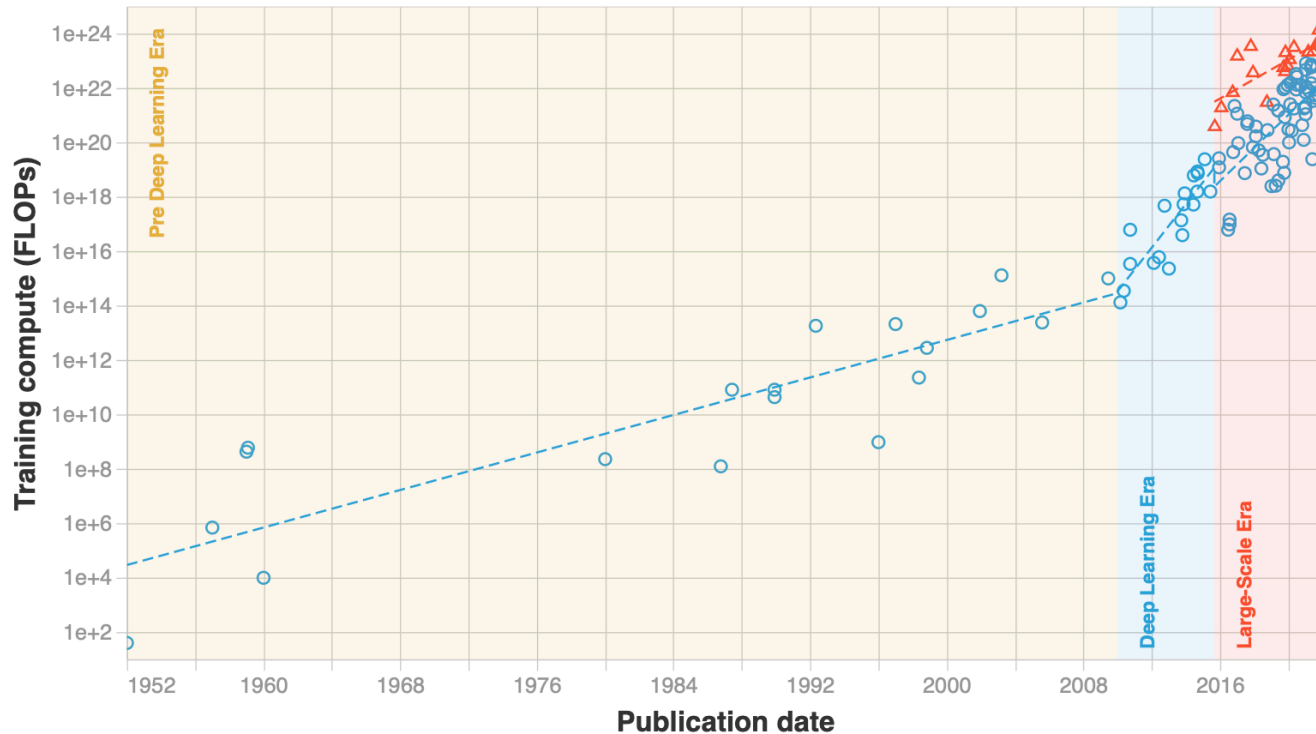
(b) 100% train, 7% test

Huang et al., 2020

Computational Cost

Training compute (FLOPs) of milestone Machine Learning systems over time

n = 121



Sevilla et al., 2022

Neural Network Training

Basis: $d\theta = G(\theta) dt$, with neural network parameters $\theta \in \mathbb{R}^n$
and G the negative gradient of the loss of entire dataset.

SGD: train on randomly selected subset of training data
 $\Rightarrow$ replace pure gradient by noisy gradient $\tilde{G}(\theta)$.

Neural Network Training

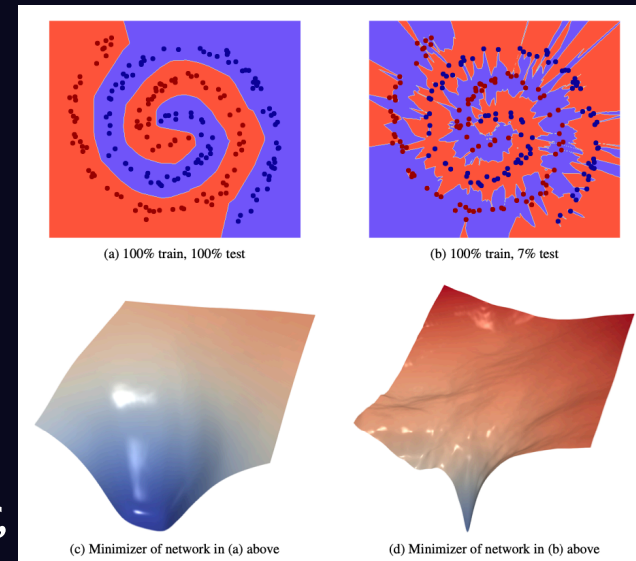
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- Implicit bias.
- Regularization.
- Structure of neural network loss landscapes.
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*Many minima: same training error,
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Further extension: incorporate momentum p .

Neural Network Training

Underdamped Langevin Dynamics framework:

$$d\theta = p \, dt$$

$$dp = \tilde{G}(\theta) \, dt - \gamma p \, dt + \sqrt{2\gamma\tau} \, dW$$

In ideal setting can show to be ergodic, i.e., solutions of the dynamics sample $\rho_\tau \propto e^{-H/\tau}$.

Limit $\tau = 0$ gives a form of SGD with momentum, whereas $\gamma \rightarrow \infty$ gives SGLD.

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Langevin Dynamics Discretization

Overdamped Langevin Dynamics: $d\theta = G(\theta) dt + \sqrt{2\tau} dW$

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Langevin Dynamics Discretization

Overdamped Langevin Dynamics: $d\theta = G(\theta) dt + \sqrt{2\tau} dW$

Euler-Maruyama discretization: $\theta_{n+1} = \theta_n + hG(\theta_n) + \sqrt{2\tau h} R_n$

with timestep h and R_n a vector of i.i.d. standard normal random variables

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Splitting methods:

$$\mathcal{U}_h^A(\theta, p) = (\theta + hp, p),$$

$$\mathcal{U}_h^B(\theta, p) = (\theta, p + hG(\theta)),$$

$$\mathcal{U}_h^O(q, p) = (\theta, e^{-\gamma h} p + \sqrt{\tau(1 - e^{-2\gamma h})} R).$$

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Different orderings possible. For example:

Algorithm 1 The OBA Splitting Scheme

```
1: procedure OBA( $\theta, p, \gamma, \tau, T, h$ )  
2:   for  $t \leftarrow 1$  to  $T$  do  
3:      $R \leftarrow \mathcal{N}(0, 1)$   
4:      $p \leftarrow \exp(-\gamma h)p + \sqrt{\tau(1 - \exp(-2\gamma h))}R$   
5:      $p \leftarrow p + hG(\theta)$   
6:      $\theta \leftarrow \theta + hp$   
7:   return  $\theta, p$ 
```

Thermodynamic Parameterization of Neural Networks

Foundations of Data Science 2019 with B. Leimkuhler and C. Matthews

Underdamped Langevin Dynamics:

$$d\theta = p \, dt$$

$$dp = G(\theta) \, dt - \gamma p \, dt + \sqrt{2\gamma\tau} \, dW$$

Hyperparameter τ controls: pure optimization ($\tau = 0$) $\leftrightarrow$ pure sampling ($\tau = 1$).

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Tempering the posterior:

$$\rho_\tau \propto e^{-H/\tau}$$

Improved performance at $\tau \ll 1$.

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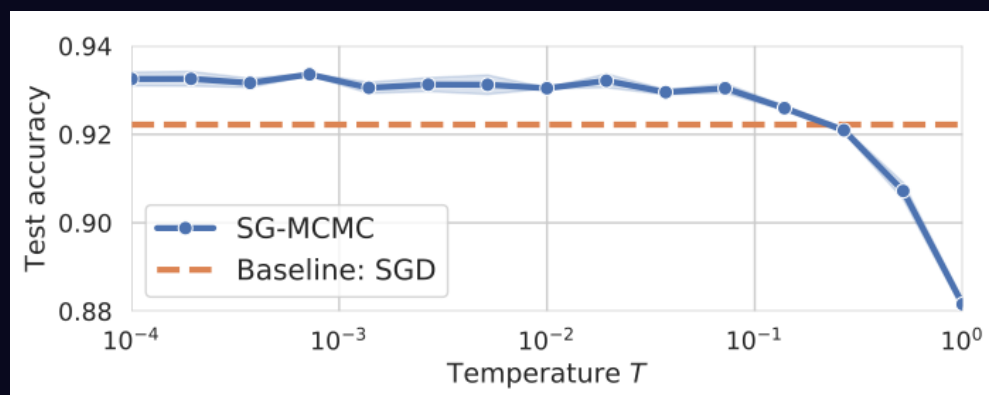
ResNet-20, CIFAR-10

Improved performance at $\tau \ll 1$.

Cold posterior effect.

Further reading:

Noci et al., NeurIPS 2021.



Wenzel et al., ICML 2020

Neural Network Training

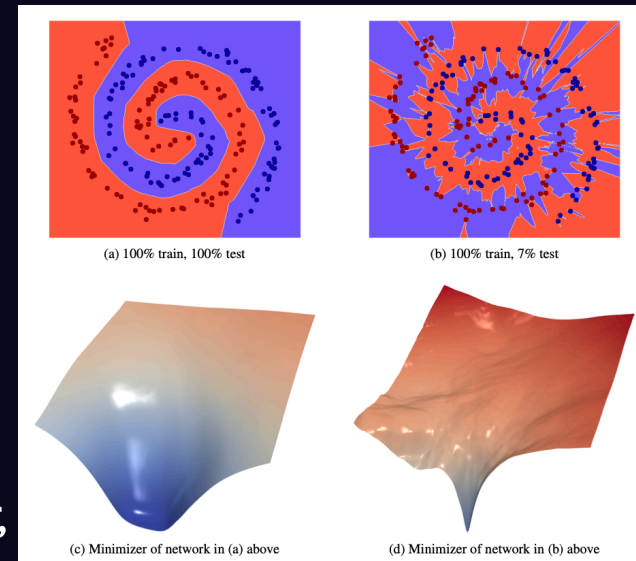
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Better Training using Weight-Constrained Stochastic Dynamics

ICML 2021

Co-authors: Benedict Leimkuhler, Timothée Pouchon, and Amos Storkey.

Constrained SDEs

Training schemes in ML:

Stochastic Gradient Descent (SGD), SGD + momentum,

Additional additive noise: Stochastic Gradient Langevin Dynamics (SGLD).

+ often L_2 Regularization / weight decay

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Stochastic Gradient Descent (SGD), SGD + momentum,

Additional additive noise: Stochastic Gradient Langevin Dynamics (SGLD).

+ often L_2 Regularization / weight decay

We consider:

- Constrained SGLD / overdamped Langevin.
- The case with momentum: Constrained underdamped Langevin.

Types of Constraints

Constraint: $g(\theta) = 0$.

For case with momentum p : Cotangent space $\nabla^T g(\theta) p = 0$.

Orthogonal Constraints: $g = W^T W - I$, weight matrix W

Circle Constraints: $g = |\theta_i^c|^2 + |\xi_i|^2 - r_i^2$, with radius r_i and slack variable ξ_i
for each constrained parameter θ_i^c

What do our training schemes look like?

Discretization of Constrained Langevin Dynamics

Initialize parameters to obey the constraint!

- Overdamped case: Euler-Maruyama step + projection back.
Best choice constraint-specific

Orthogonal constraints
where $Q = W^l$

Algorithm 1 Orthog. constraint overdamped Langevin

Every step:

$$Q^{(0)} = Q_n - h\tilde{G}(Q_n) + \sqrt{2\tau h}R_n,$$

for $k = 0$ **to** $K - 1$ **do**

$$Q^{(k+1)} = Q^{(k)} - \frac{1}{2}Q_n((Q^{(k)})^T Q^{(k)} - I_s),$$

end for

$$Q_{n+1} = Q^{(K)}.$$

Discretization of Constrained Langevin Dynamics

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- Underdamped case: Constrained analogue of splitting schemes

Orthogonal constraints
Projection onto
cotangent space

$$\bar{P} \mapsto \Pi_Q \bar{P} = \bar{P} - \frac{1}{2}Q(\bar{P}^T Q + Q^T \bar{P}).$$

Algorithm 2 Orthog. constraint underdamped Langevin

Every step:

$$Q^{(0)} = Q_n + hP_n,$$

for $k = 0$ **to** $K - 1$ **do**

$$Q^{(k+1)} = Q^{(k)} - \frac{1}{2}Q_n((Q^{(k)})^T Q^{(k)} - I_s),$$

end for

$$Q_{n+1} = Q^{(K)}, \bar{P}_{n+1} = P_n + \frac{1}{h}(Q_{n+1} - Q^{(0)}),$$

$$P_{n+1} = \Pi_{Q_{n+1}} \bar{P}_{n+1}$$

} (A)

$$\bar{P}_{n+1} = P_n - h\tilde{G}(Q_n),$$

$$P_{n+1} = \Pi_{Q_n} \bar{P}_{n+1},$$

} (B)

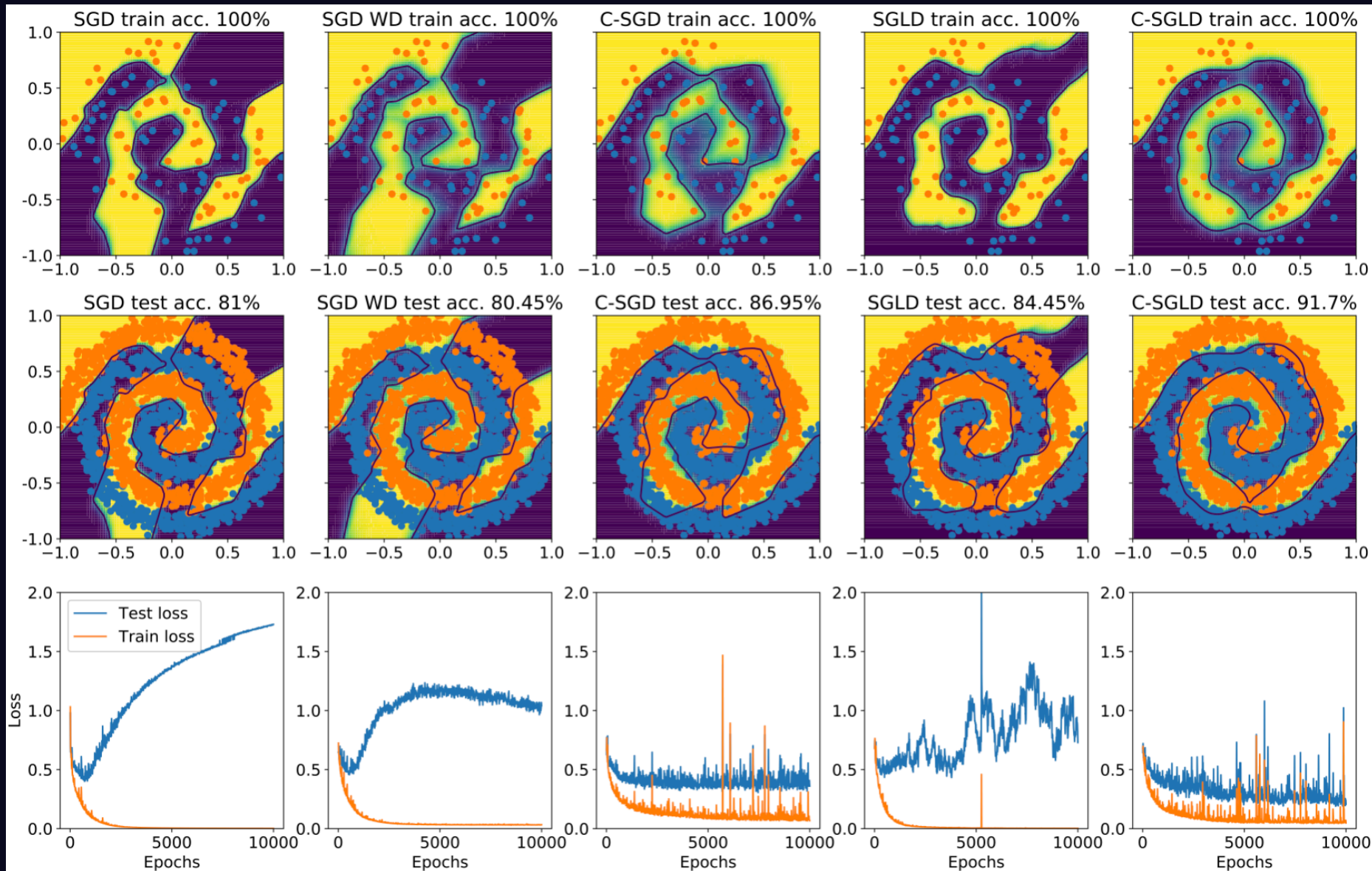
$$P_{n+1} = e^{-\gamma h}P_n + \sqrt{\tau(1 - e^{-2\gamma h})}R_n,$$

$$P_{n+1} = \Pi_{Q_n} \bar{P}_{n+1}$$

} (O)

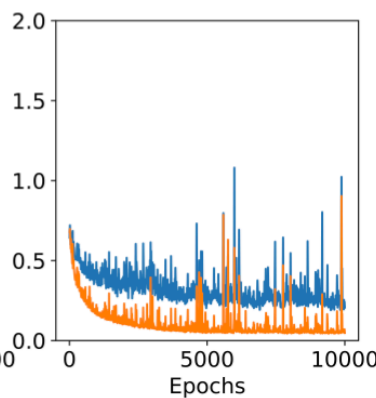
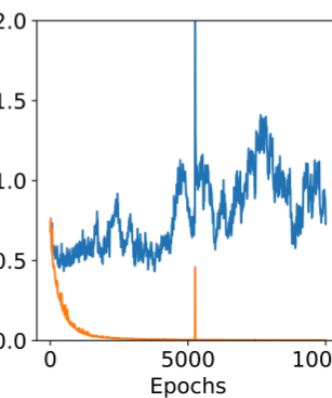
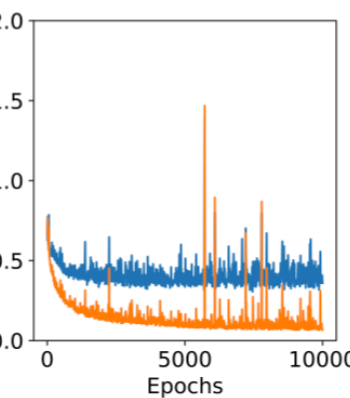
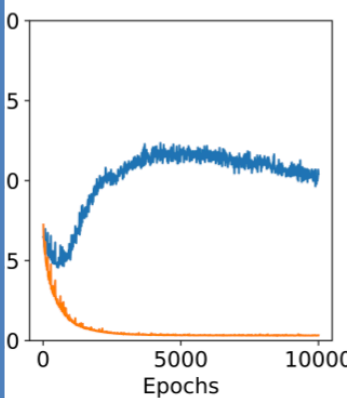
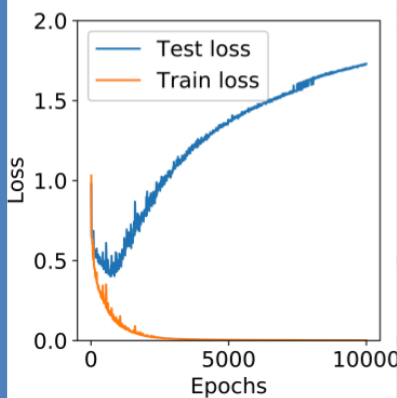
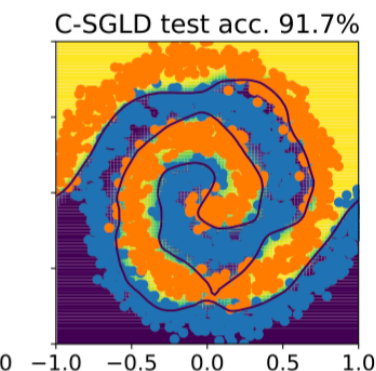
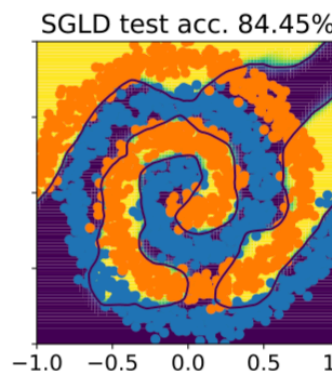
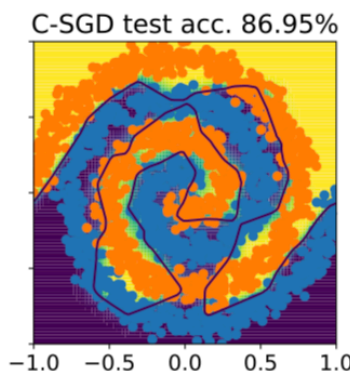
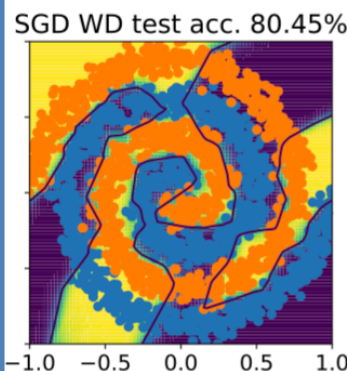
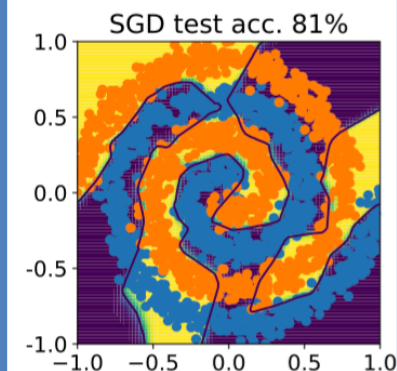
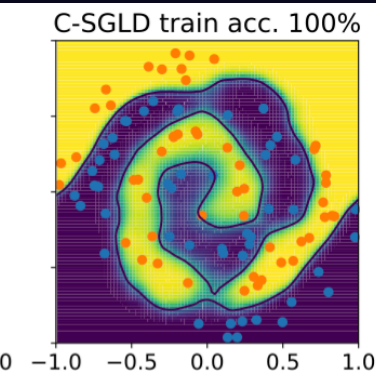
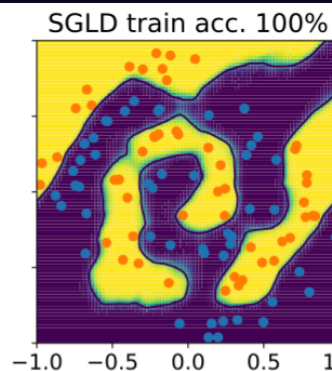
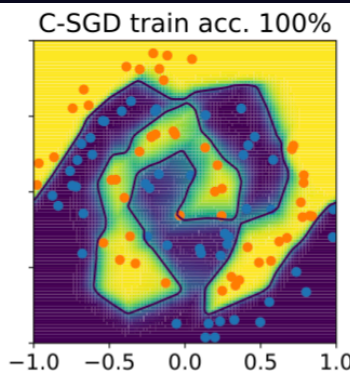
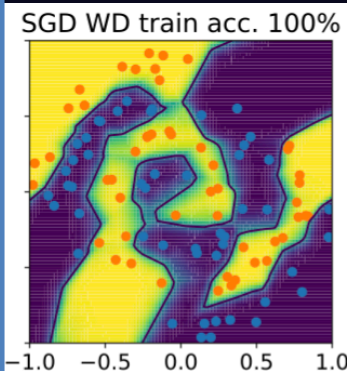
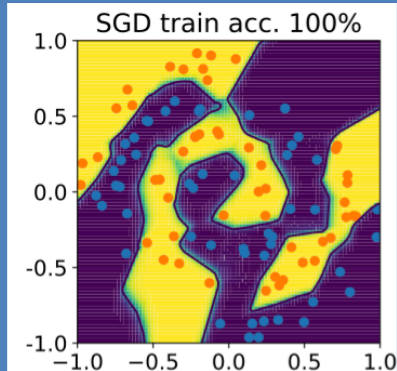
Hard Constraints

Soft Constraint



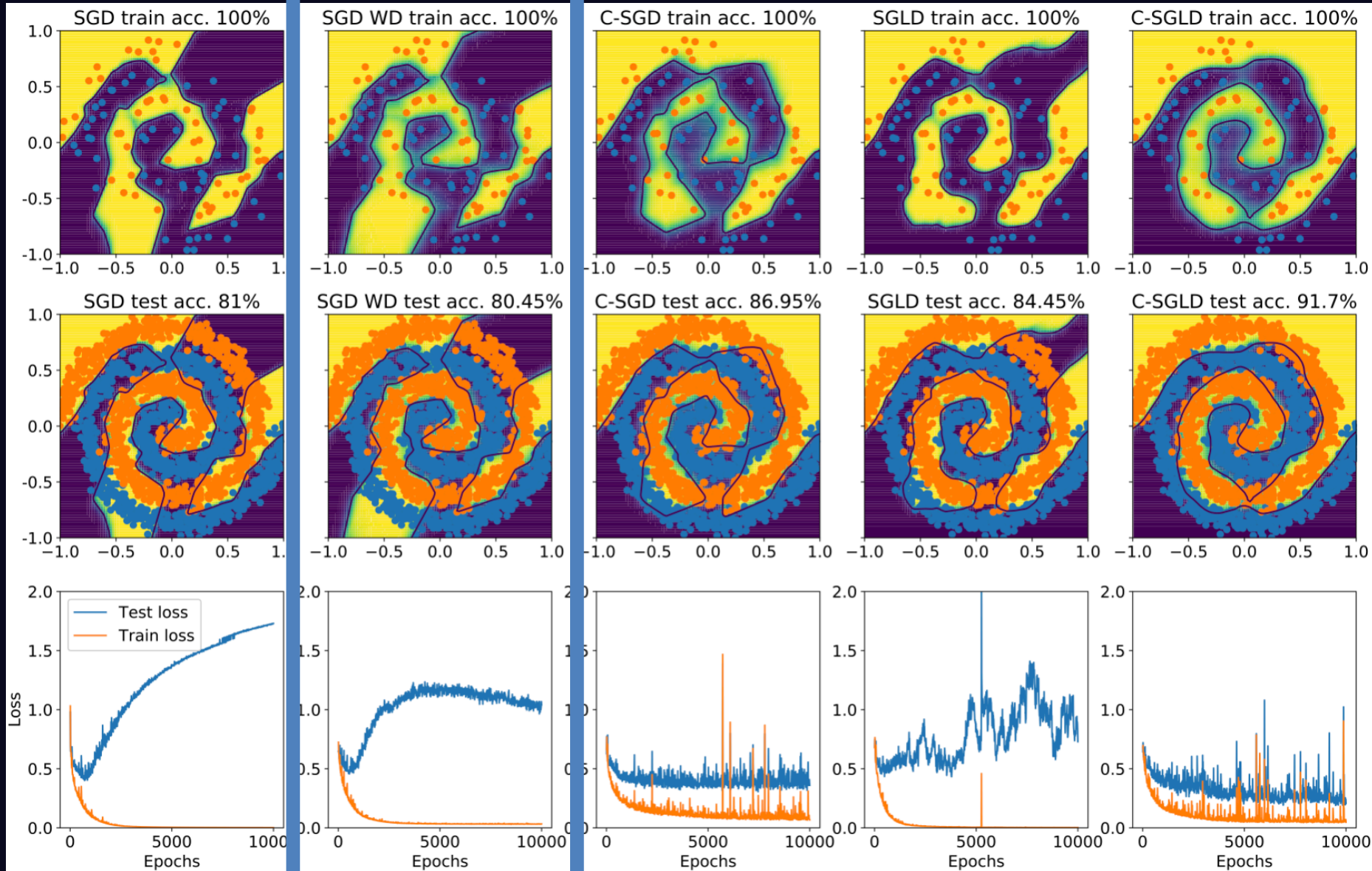
Hard Constraints

Soft Constraint

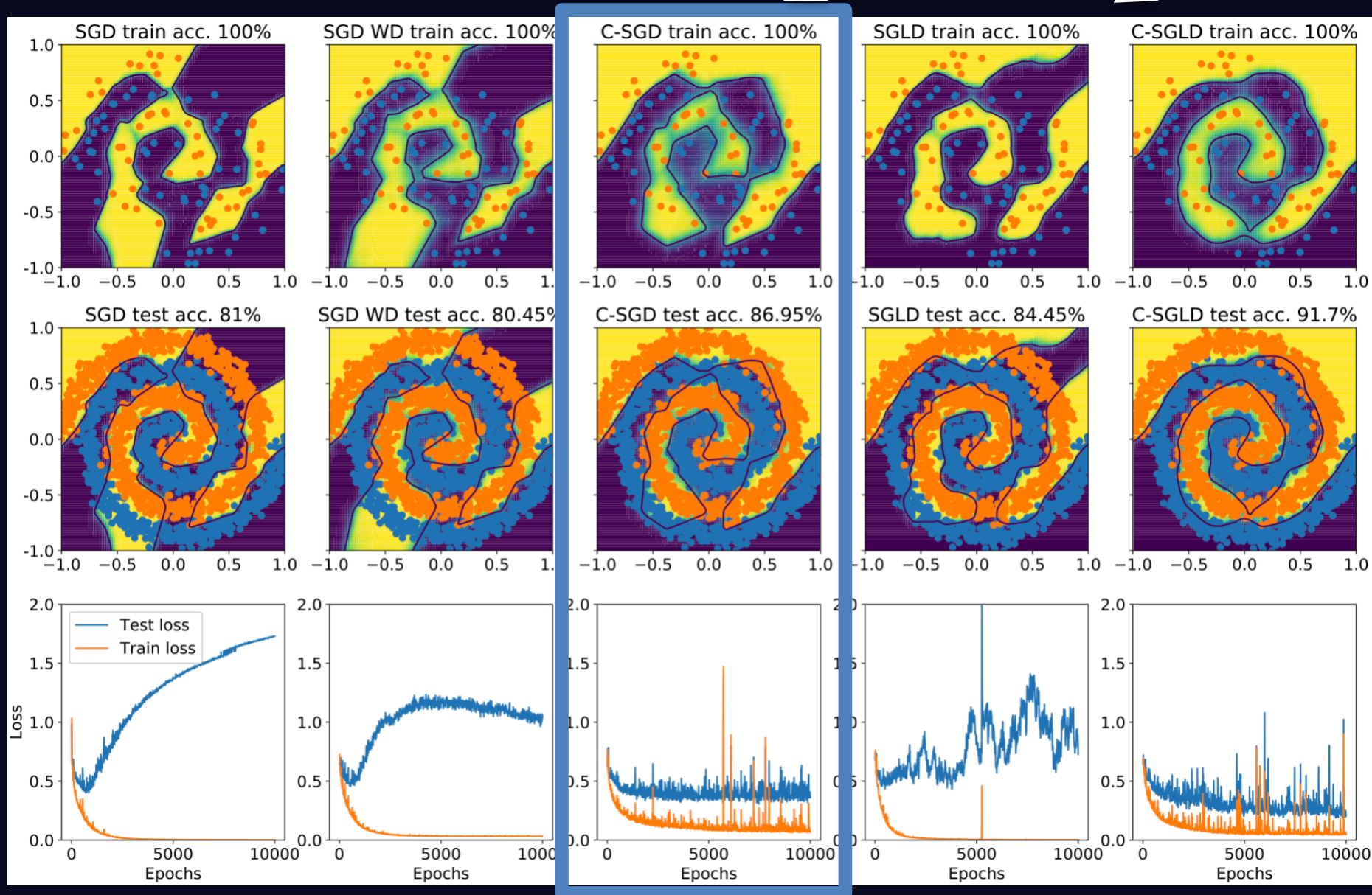


Hard Constraints

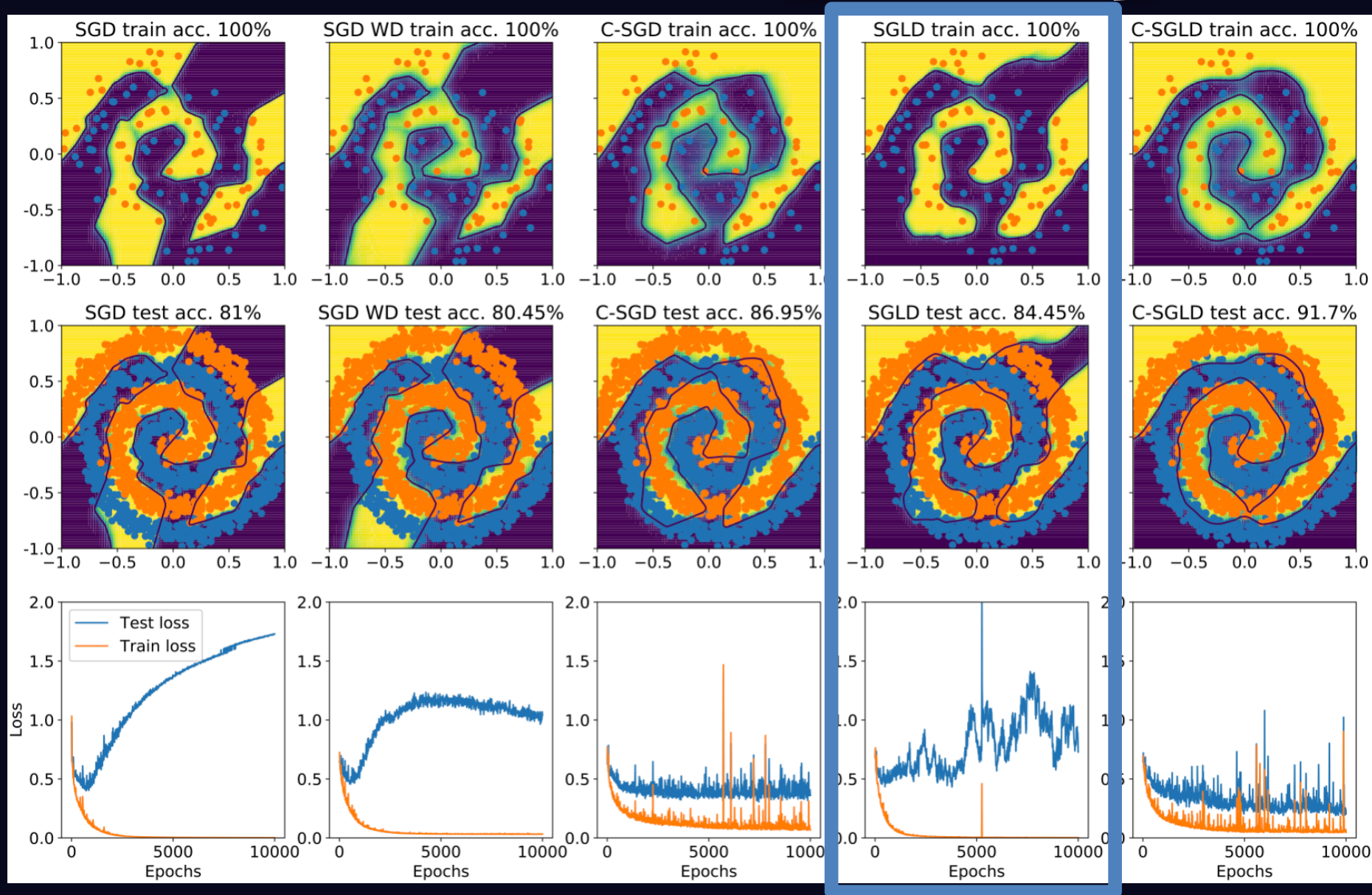
Soft Constraint



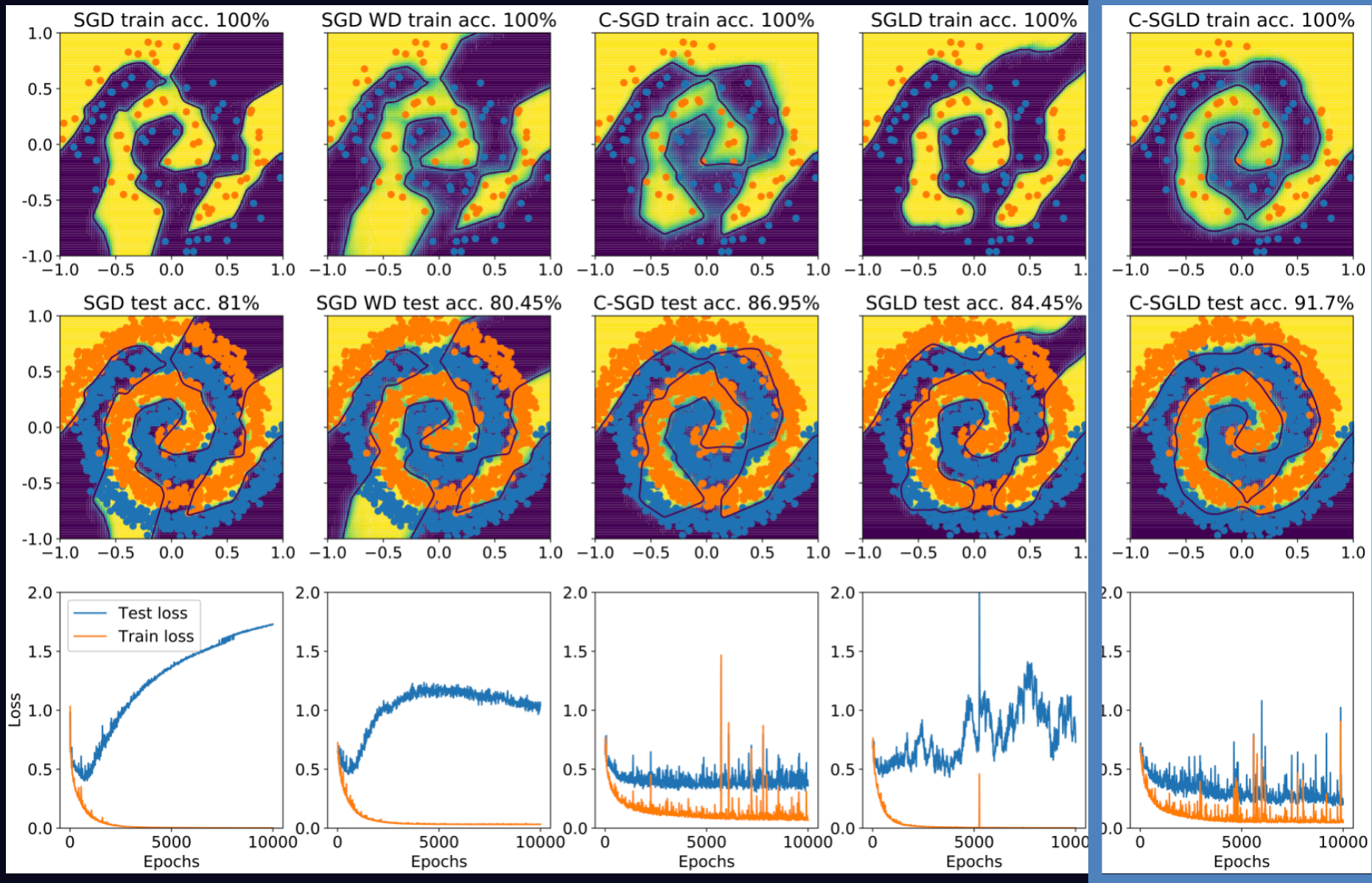
Hard Constraints



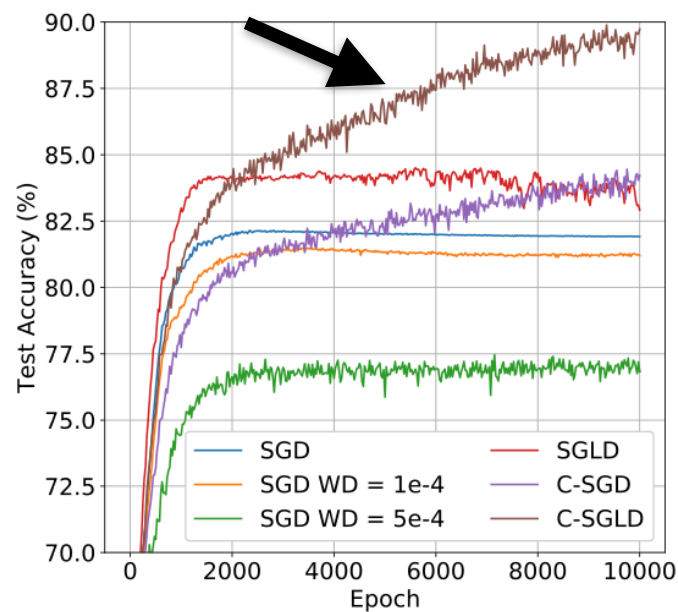
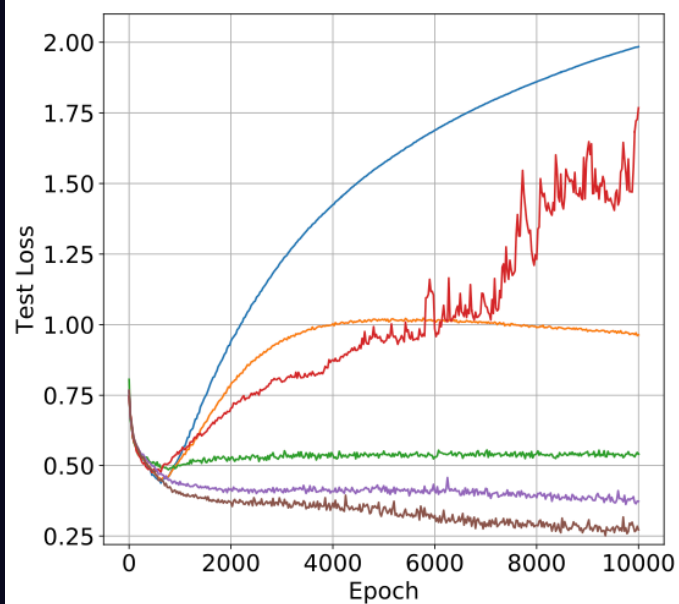
Hard Constraints



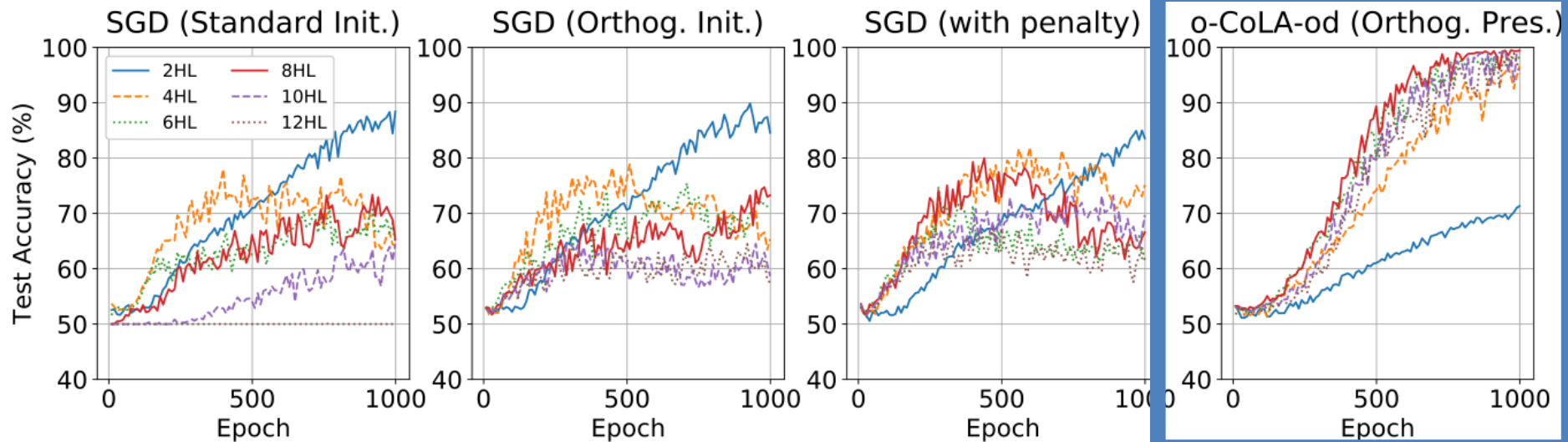
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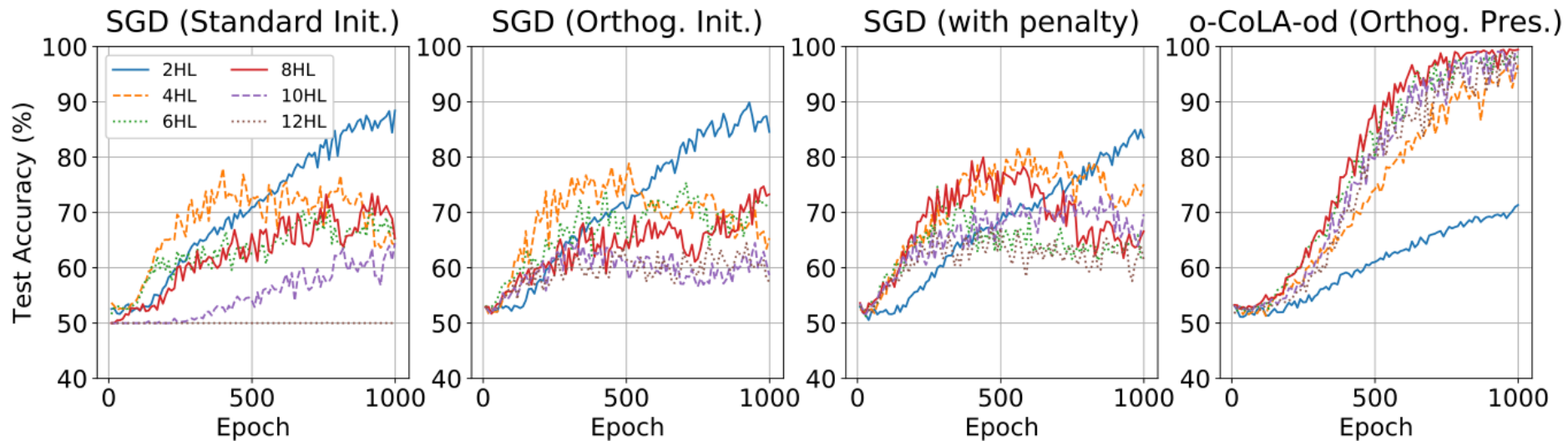
Over 100 runs..
Same effect visible



Orthogonal Constraints



Orthogonal Constraints



See paper for: ResNet-34, NLP, Fashion-MNIST, soft constraints comparisons

Neural Network Training

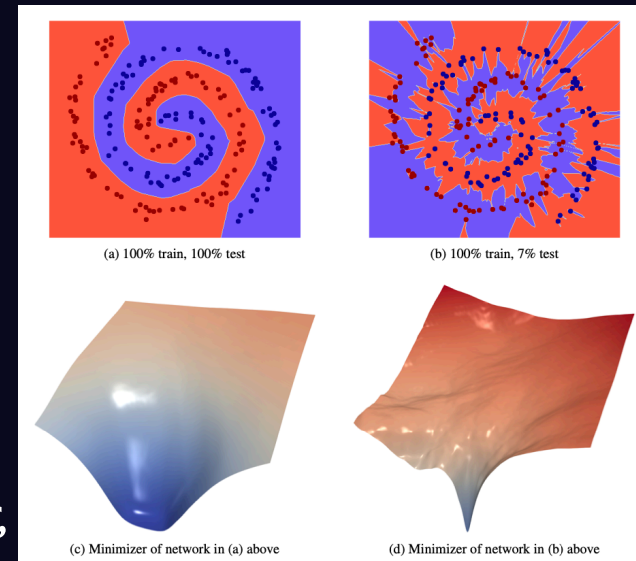
Basis: $d\theta = G(\theta) dt$, with neural network parameters $\theta \in \mathbb{R}^n$
and G the negative gradient of the loss of entire dataset.

SGD: train on randomly selected subset of training data
 $\Rightarrow$ replace pure gradient by noisy gradient $\tilde{G}(\theta)$.

Many open questions on:

- Implicit bias.
- Regularization.
- Structure of neural network loss landscapes.
-

*Many minima: same training error,
different generalization performance.*



Linear Interpolation of Neural Network Loss Landscapes

$$\theta_\alpha = (1 - \alpha)\theta_i + \alpha\theta_f \text{ for } \alpha \in [0, 1]$$

Goodfellow et al., 2015

↑ ↑
Initial Final Model Parameters

Linear interpolation is seen as “a simple and lightweight method to probe neural network loss landscapes” (Lucas et al., 2021).

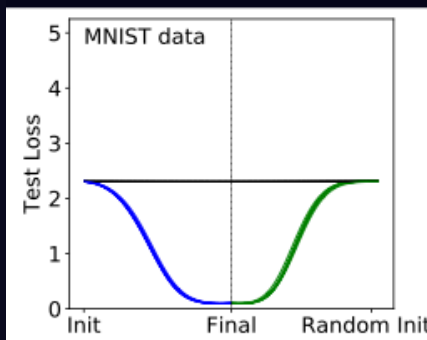
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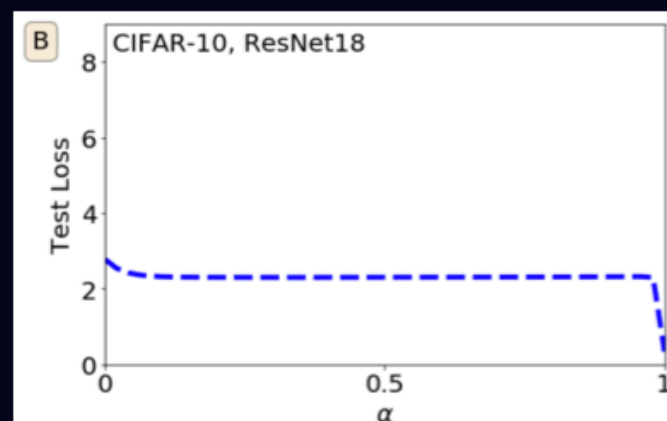
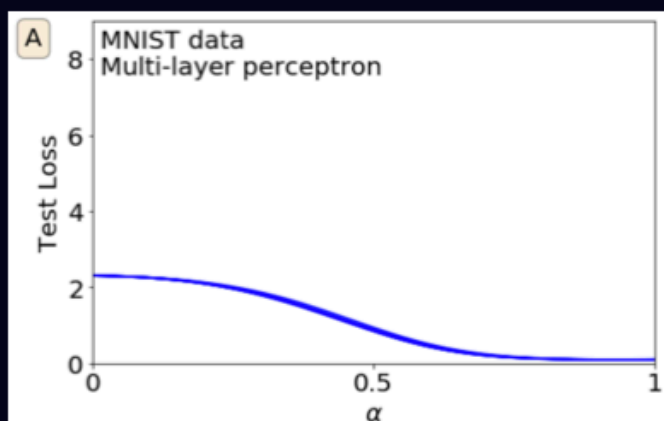
Goodfellow et al., 2015

An absence of barriers along the linear path

⇒ “tasks are relatively easy to optimize” (Goodfellow et al., 2015).

⇒ “Though dimension is high, the space is in some sense simpler than we thought: [...] the walk could just as well have taken a straight line without encountering any obstacles” (Li et al., 2018).

Revisited: Linear Interpolation of Neural Network Loss Landscapes



In modern neural network architectures:
“Loss plateaus and error remains at the level of random chance ...
... until near the optimum” (Frankle, 2020).

Similar observations by Lucas et al. (2021).

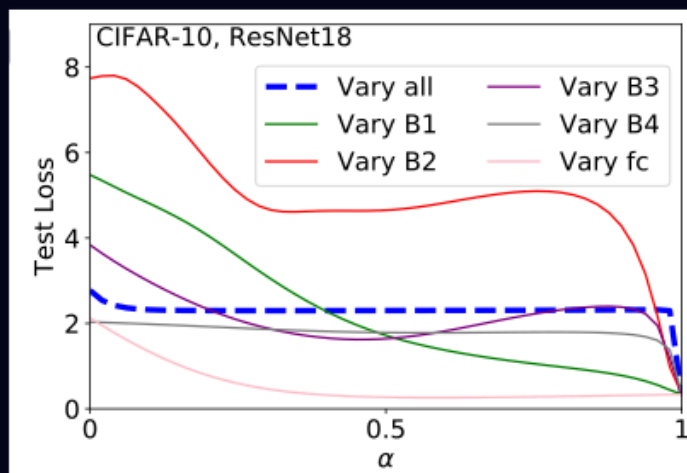
Layer-wise Linear Interpolation

Vary a single layer (or convolutional block) from initial to final state

$$\theta_{\alpha}^{(\ell)} = (1 - \alpha)\theta_0^{(\ell)} + \alpha\theta_f^{(\ell)}, \quad \theta_{\alpha}^{(k)} = \theta_f^{(k)}, \quad k \neq \ell.$$

Chatterji et al., 2020

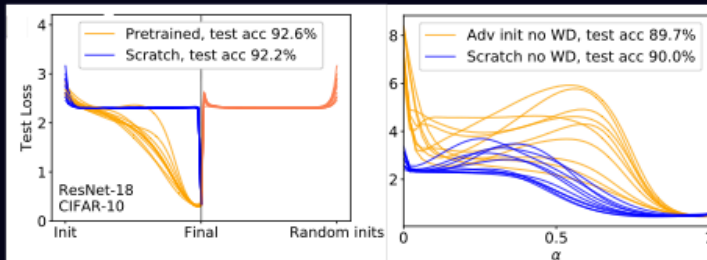
Keep all other parameters fixed at their final state.



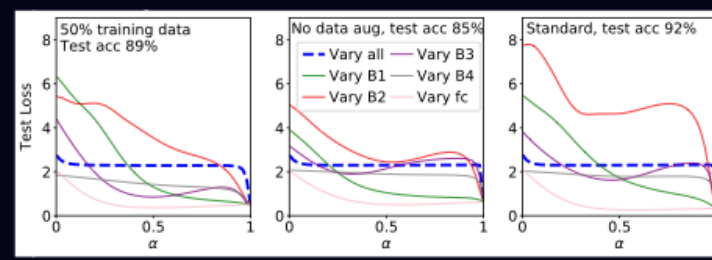
Interventions

Effect on: 1) Shape of Linear Path 2) Final Test Accuracy

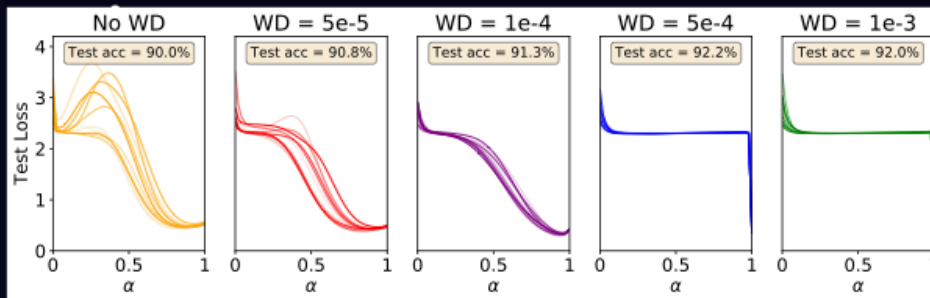
Role of Initialization



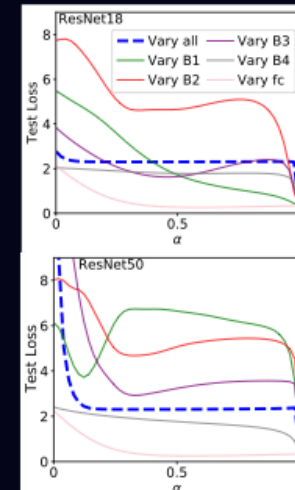
Role of Data



Role of Optimization



Role of Model



Findings

Question I: Does the shape of loss along the linear path relate to the “success” of optimization?

⇒ No!

An absence of barriers along the linear path.. “tasks are relatively easy to optimize” (Goodfellow et al., 2015).

Question II: Does the shape of loss along the linear path relate to other aspects of optimization?

⇒ Pre-training on ImageNet consistently removes the presence of barriers for ResNet architectures, whereas adversarial initialization on random labels increases barriers.

“pre-trained weights guide the optimization to a flat basin of the loss landscape.” (Neyshabur et al., 2020)

⇒ Distance between initial and final parameter state is not a reliable indicator of non-monotonic behaviour along linear path.

“Large distances moved in weight space encourage non-monotonic interpolation” (Lucas et al., 2021).

Layer-wise Findings

The adversarial effect of partial pre-training

Set model to trained (T) state

- Re-set specific layer to random initialization (RI)
- Re-train
- Worse test accuracy

ResNet-18, CIFAR-10

Method	Test acc (%)
T-All but RI-1	91.79 $\pm$ 0.23
T-All but RI-2	91.83 $\pm$ 0.21
T-All but RI-3	92.35 $\pm$ 0.20
T-All but RI-4	90.97 $\pm$ 0.31
Train from scratch	92.2%

Background: Loss Landscapes

Normalization layers are thought to reduce sharpness [Santurkar et al., 2018, Bjorck et al., 2018, Lyu et al., 2022].

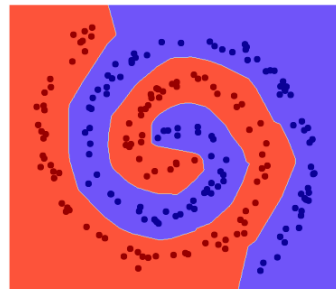
What is sharpness?

Background: Loss Landscapes

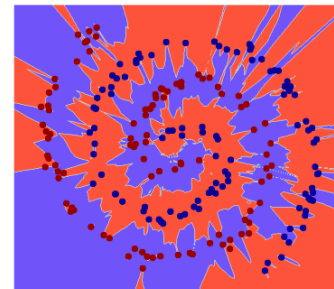
Normalization layers are thought to reduce sharpness [Santurkar et al., 2018, Bjorck et al., 2018, Lyu et al., 2022].

What is sharpness?

Recall many minima: same training error, different generalization performance.



(a) 100% train, 100% test

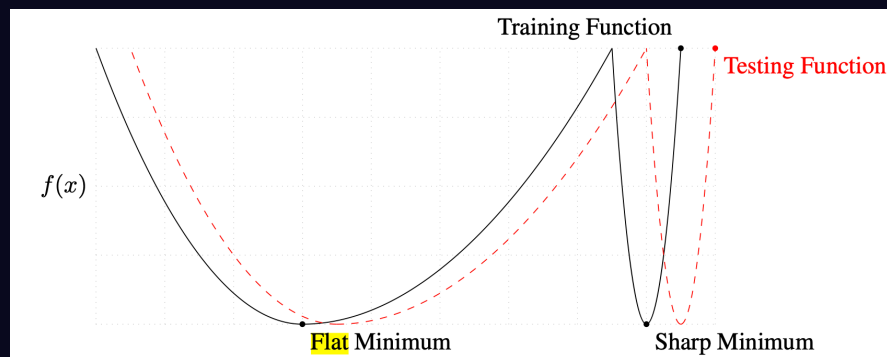


(b) 100% train, 7% test

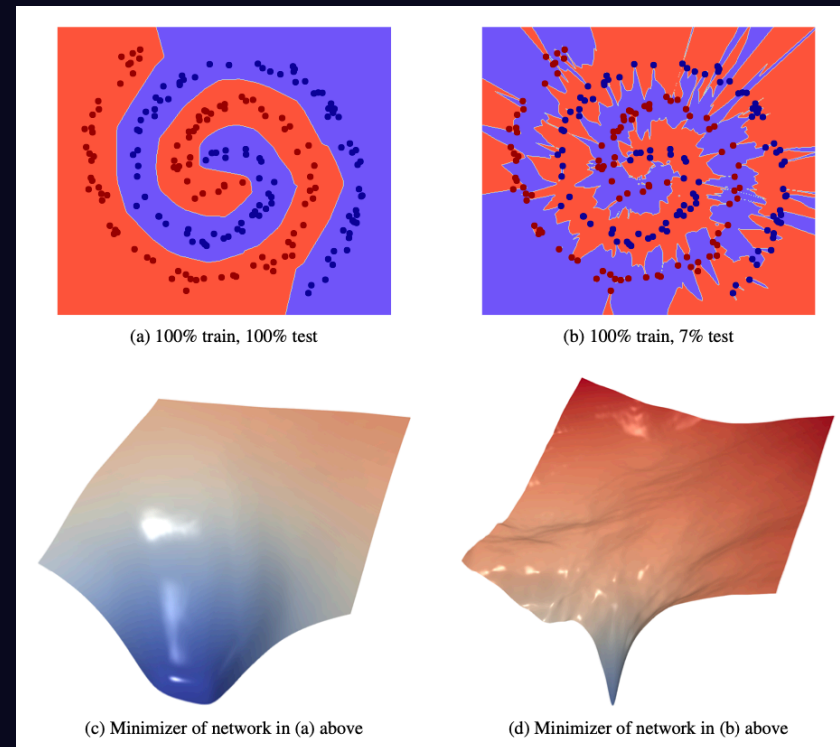
Huang et al., 2020

Background: Flatness and Generalization

Long history. Hochreiter and Schmidhuber, 1997.



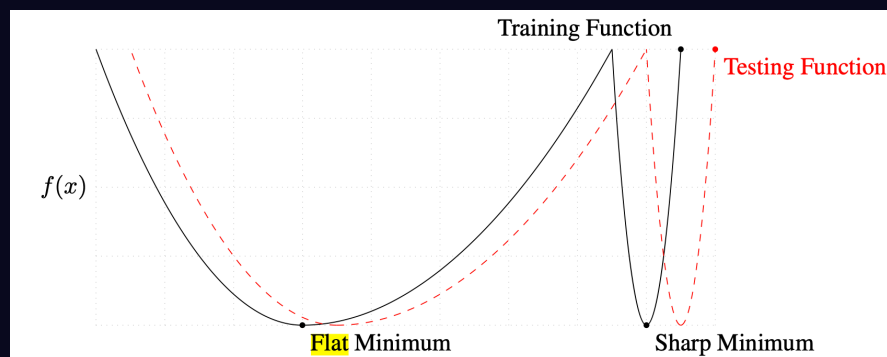
“Since flat minimizers can be specified with lower precision than sharp minimizers, they tend to have better generalization performance”, Keskar et al., 2017.



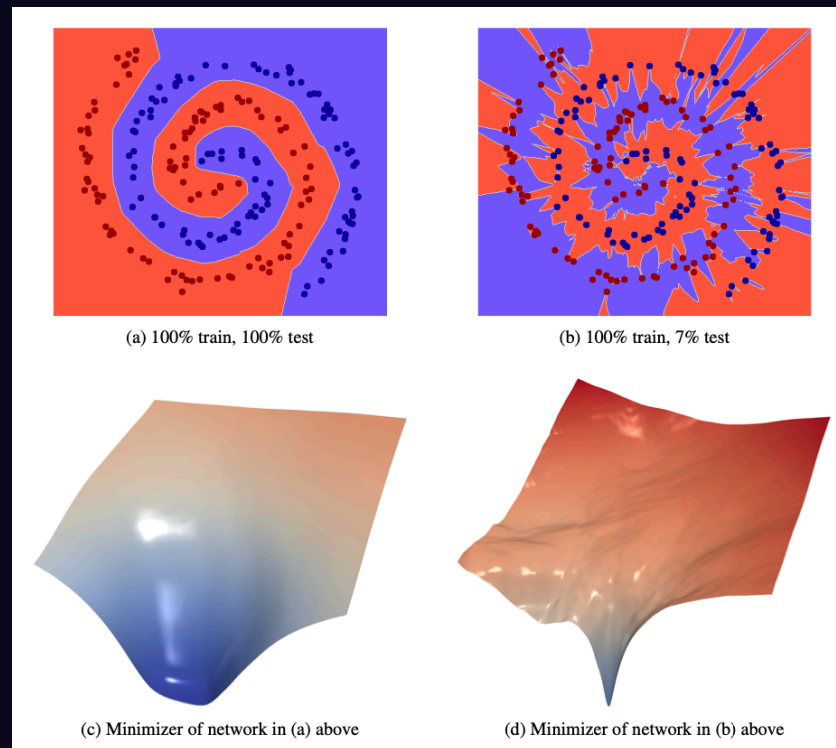
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Background: Flatness and Generalization

Long history. Hochreiter and Schmidhuber, 1997.



“Since flat minimizers can be specified with lower precision than sharp minimizers, they tend to have better generalization performance”, Keskar et al., 2017.



Huang et al., 2020

Some critiques... e.g., Dinh et al., 2017, Andriushchenko et al. 2023.

“sharper minima can generalize better”.

Sharpness-Aware Minimization (SAM)

Foret et al., 2021

Change the objective. Find minima that have both low loss and low sharpness.

$$\min_w \max_{\|\epsilon\|_2 < \rho} L(w + \epsilon)$$

Sharpness-Aware Minimization (SAM)

Foret et al., 2021

Change the objective. Find minima that have both low loss and low sharpness.

$$\min_w \max_{\|\epsilon\|_2 \leq \rho} L(w + \epsilon) \quad \max_{\|\epsilon\|_2 \leq \rho} L(w + \epsilon) - L(w) + L(w)$$

Sharpness-Aware Minimization (SAM)

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Additional forward-backward pass.

Sharpness-Aware Minimization (SAM)

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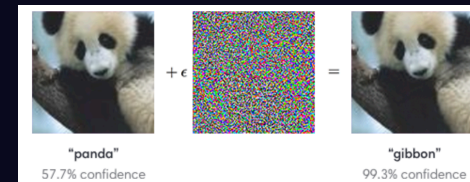
$$\min_w \max_{\|\epsilon\|_2 < \rho} L(w + \epsilon)$$

Additional forward-backward pass.

Works remarkably well !

e.g. SGD + SAM can effectively train Vision Transformers (Chen et al., 2022)

Can also enhance adversarial robustness (Zhang et al., 2024)



Adam optimizer

Kingma and Ba, 2015

Stores moving averages as estimates of first moment and second raw moment of the gradient to adaptively update effective learning rate for different parameters

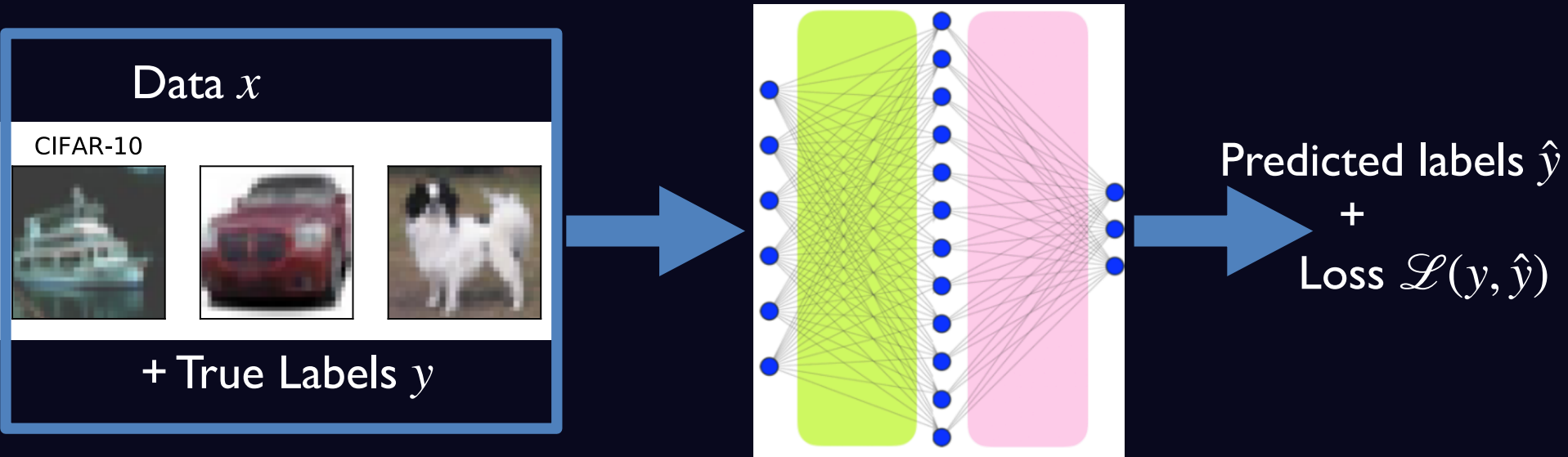
Update rule

$$\begin{aligned}m_{n+1} &= \beta_1 m_n + (1 - \beta_1) G_{(S)}(\theta_n), \\v_{n+1} &= \beta_2 v_n + (1 - \beta_2) G_{(S)}(\theta_n)^2, \\\hat{m}_{n+1} &= m_{n+1} / (1 - \beta_1^{n+1}), \\\hat{v}_{n+1} &= v_{n+1} / (1 - \beta_2^{n+1}), \\\theta_{n+1} &= \theta_n - h \hat{m}_{n+1} / (\sqrt{\hat{v}_{n+1}} + \epsilon),\end{aligned}$$

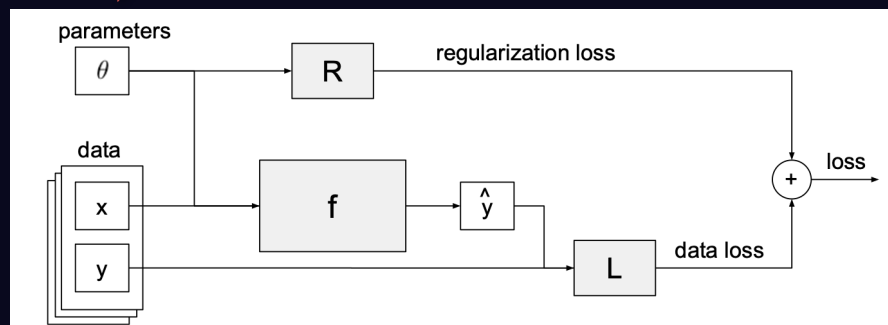
AdamW [Loshchilov and Hutter, 2019] with regularization strength β_R

$$\theta_{n+1} = \theta_n - h \left(\hat{m}_{n+1} / (\sqrt{\hat{v}_{n+1}} + \epsilon) + \beta_R \theta_n \right)$$

Framework

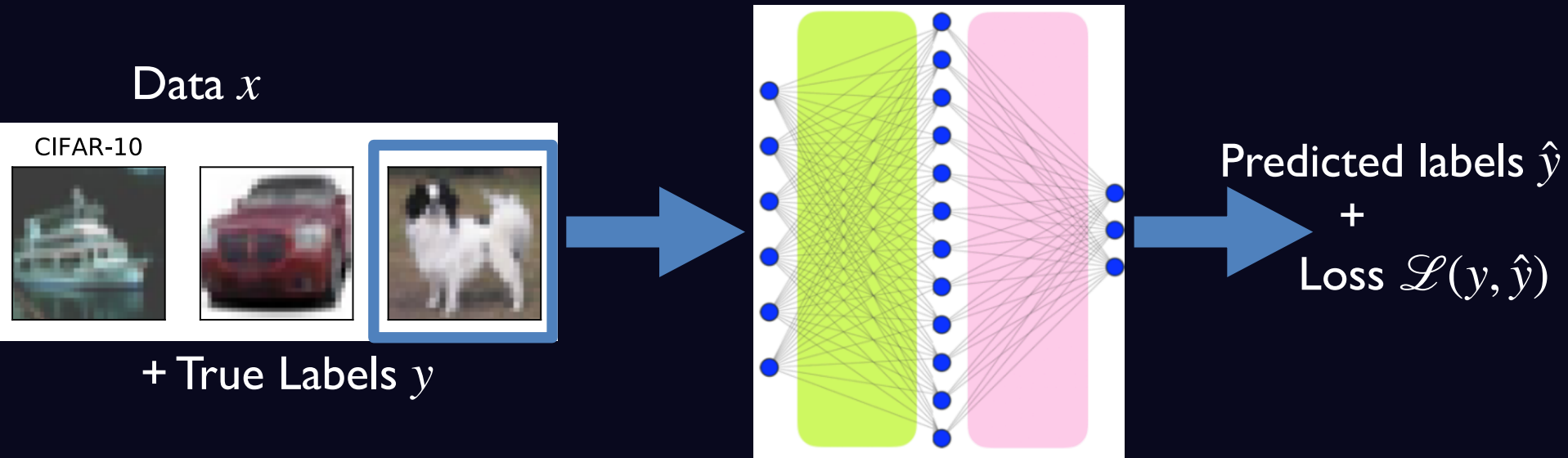


Backpropagation

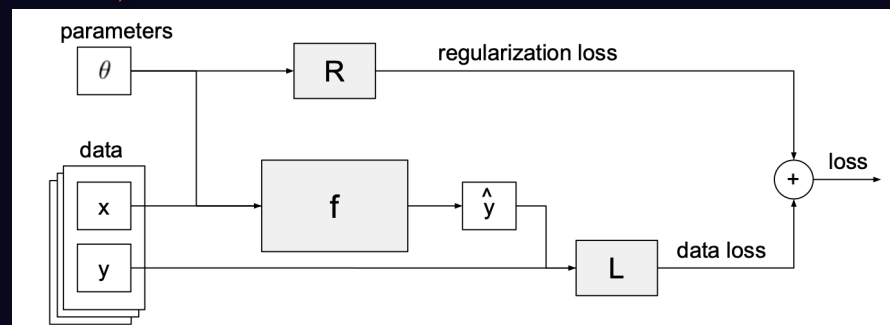


Karpathy, 2016

Framework



Backpropagation



Karpathy, 2016

Reflections on data

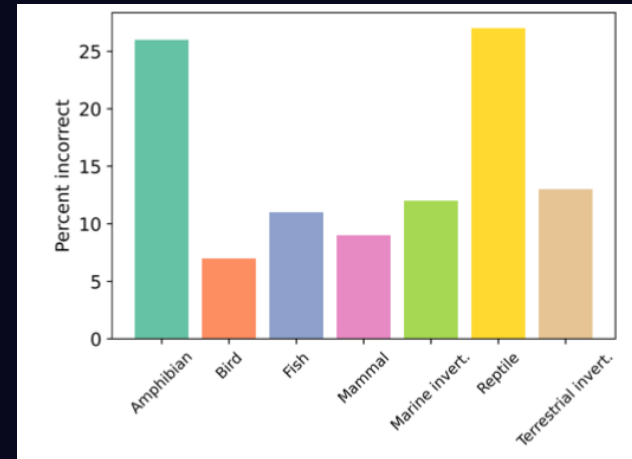
Reflections on data

ImageNet: 269 classes of wild animals [Luccioni and Rolnick, 2022].

Reflections on data

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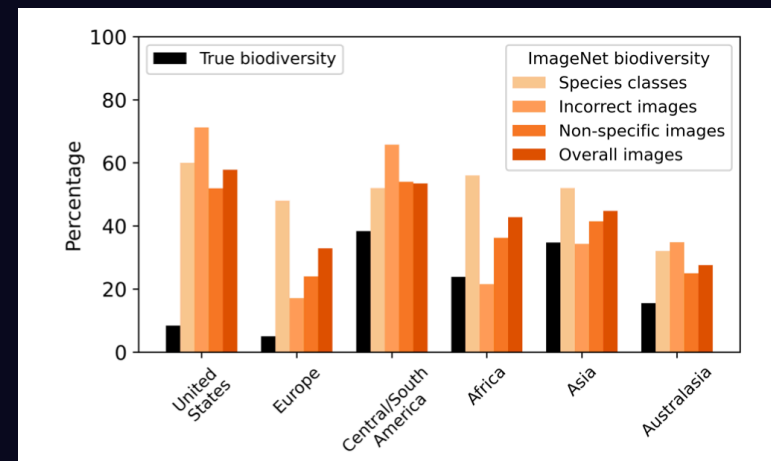
- 12% incorrectly labelled (some classes >90% incorrect labels!)
- Classes not clearly defined or overlap.



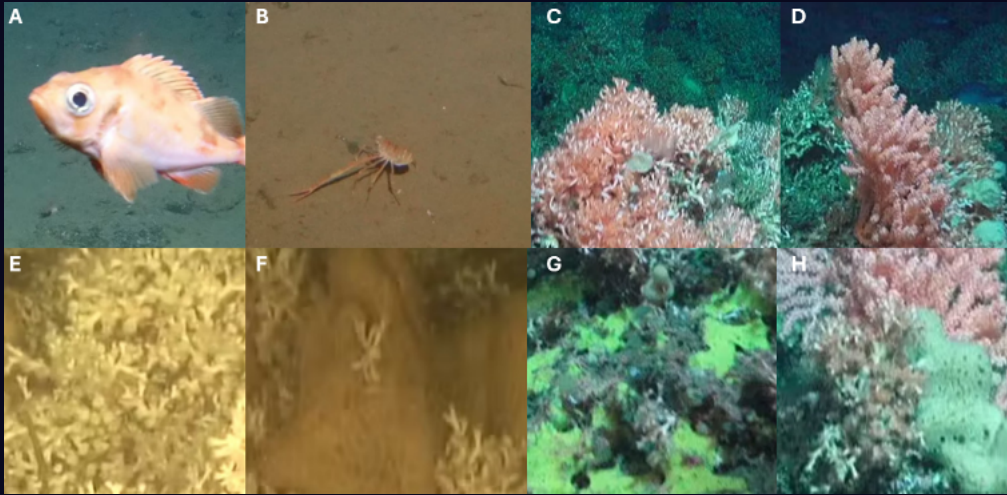
Reflections on data

ImageNet: 269 classes of wild animals [Luccioni and Rolnick, 2022].

- 12% incorrectly labelled (some classes >90% incorrect labels!)
- Classes not clearly defined or overlap.
- Biases in image choice, e.g. fish held by human vs. natural environment.
- Cultural bias, e.g. 93 mammal vs. 27 insect classes.
- Limited geographical diversity.



Real-world ecological data

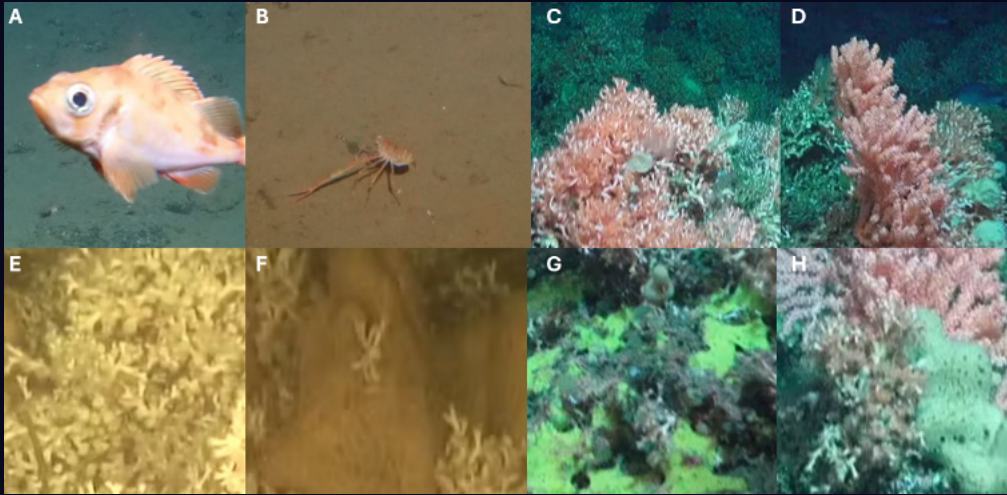


De Clippele et al. (Royal Society Open Science, 2026)

Various challenges:

- Large volumes of data.
- Underwater image quality highly variable.
- Benthic environments require extensive taxonomic expertise for annotation.

Real-world ecological data

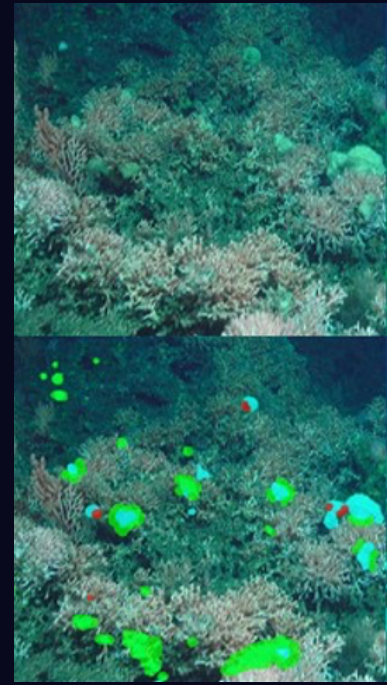


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Various challenges:

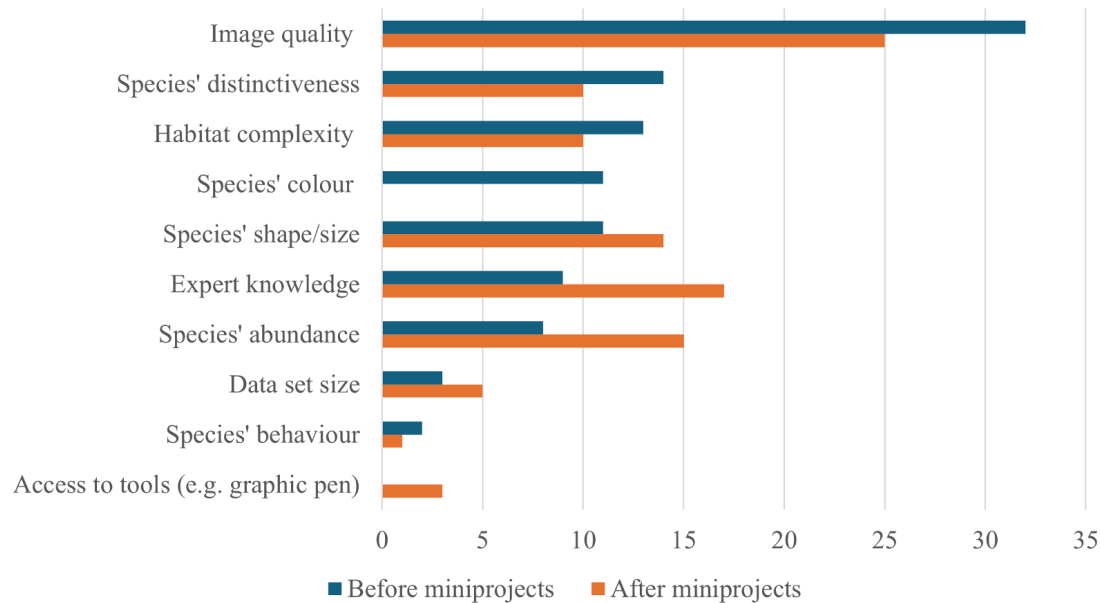
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Annotation quality impacts model performance.



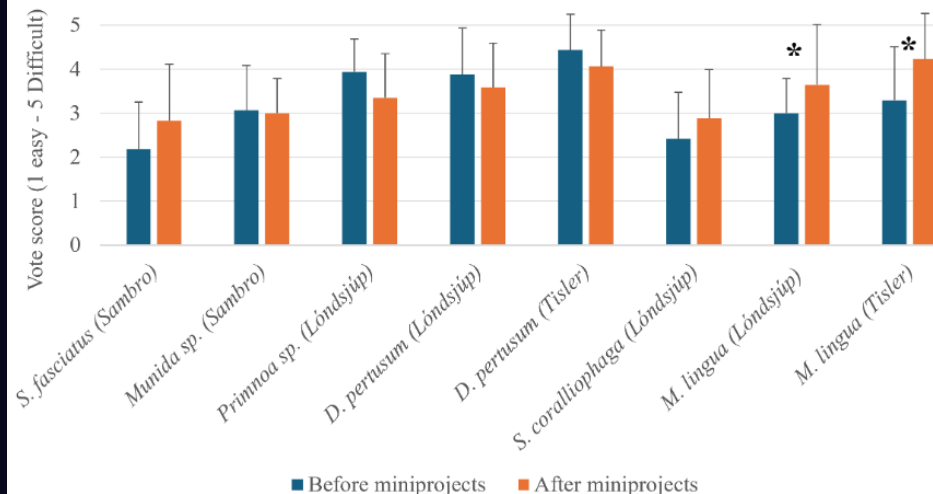
'Expert' Reflections on Example Difficulty

What affects training time and effort?



Expectations
did not align with reality

How quick/easy is it to train a model for a certain target species?

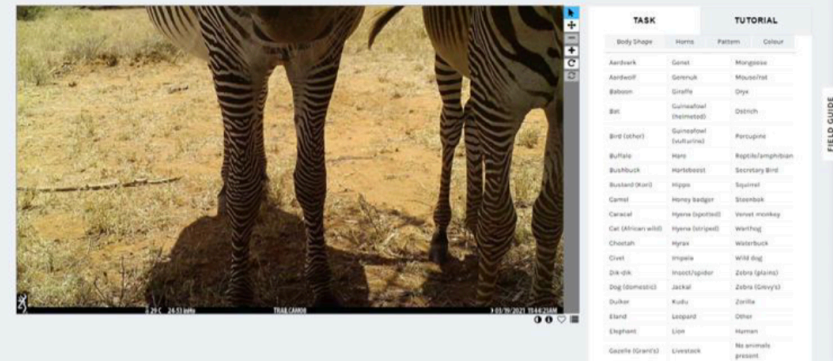


De Clippele et al.

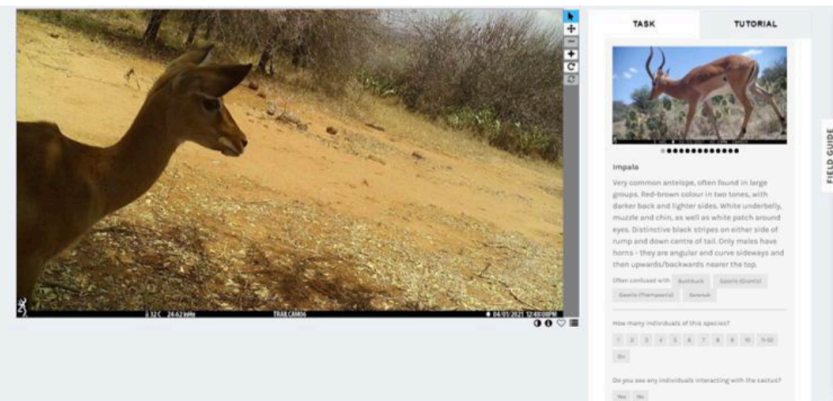
Royal Society Open Science, 2026

Citizen Science Classifications

A)



B)



Example of Zooniverse interface

Citizen Science Classifications



Citizen Science Classifications



Volunteer classifications:

1 x Elephant, 1 x Impala

Citizen Science Classifications



Volunteer classifications:

1 x Elephant, 1 x Impala



Citizen Science Classifications



Volunteer classifications:
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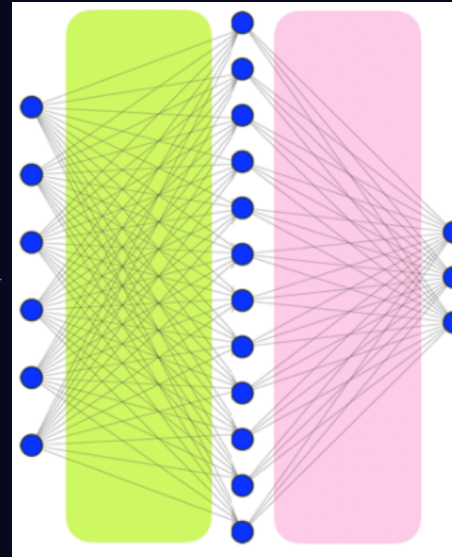


Volunteer classifications:
8x Giraffe, 2x Dikdik, 1x Camel, 1x Other

Label Ground Truth Uncertainty

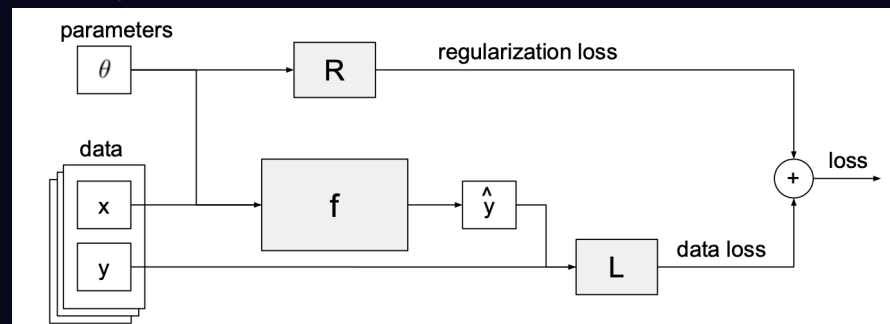


Prickly Pear Project Kenya Dataset
(Stewart et al., 2024)



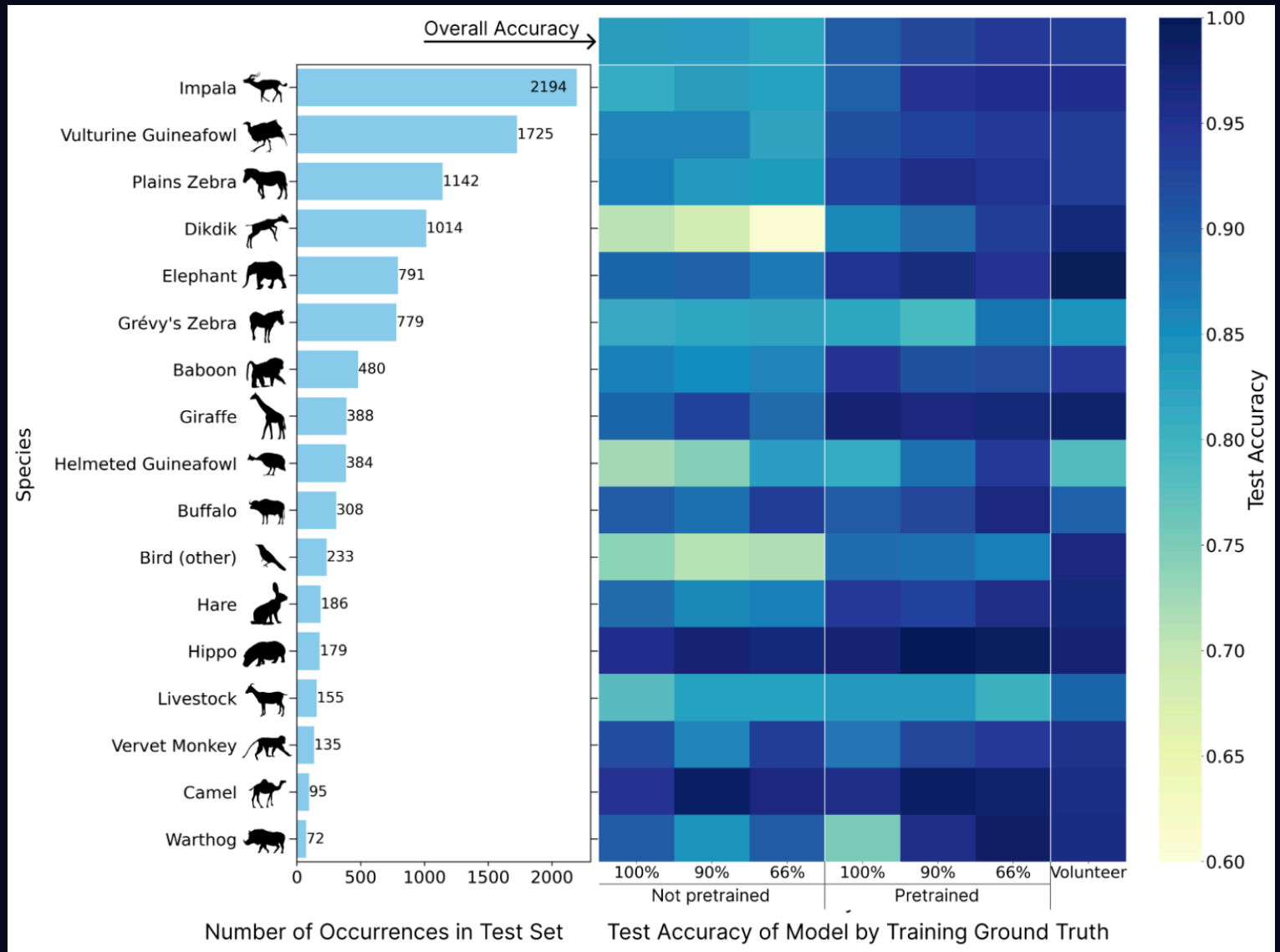
Predicted labels $\hat{y}$
+
Loss $\mathcal{L}(y, \hat{y})$

Backpropagation

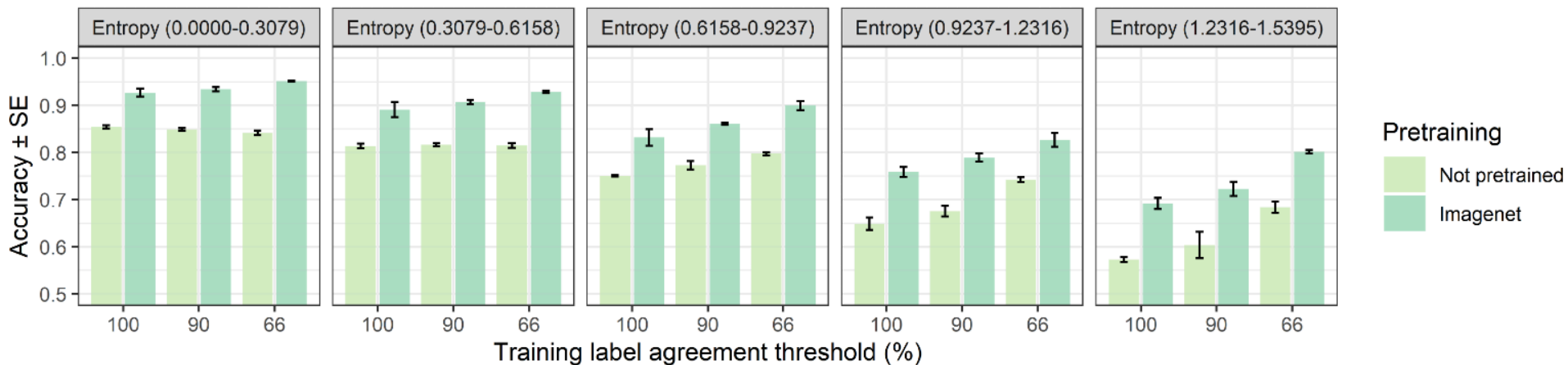


Karpathy, 2016

Training under different levels of ground truth uncertainty
 $\geq 66\%$, 90 %, and 100% volunteer agreement for label acceptance



Reflections on Example Difficulty



ResNet-50 on images with different levels of volunteer disagreement (Shannon entropy)

Questions?